

BIO5105 FILES MCQS + CONCEPTUAL Q/A + SUMMARIZED NOTES + MODEL PAST PAPER BY VULMS MASTERY

1. Which statement best defines a cell?
A) A non-living chemical particle
B) The basic structural and functional unit of life ✓
C) A tissue responsible for digestion
D) A membrane-bound organ only
2. Which statement about cell theory is correct?
A) Cells arise spontaneously from air
B) All organisms are composed of cells ✓
C) Only plants are made of cells
D) Cells do not contain genetic material
3. Which statement best explains why cells are considered functional units of life?
A) They perform essential life processes ✓
B) They exist only in animals
C) They cannot reproduce
D) They contain only water
4. Which statement correctly describes the composition of matter in living organisms?
A) Living matter is composed entirely of metals
B) Cells are mainly composed of chemical elements and compounds ✓
C) Matter in cells has no organization
D) Only carbon exists in cells
5. Which statement about atoms is correct?
A) Atoms cannot combine with each other
B) Atoms are the smallest functional units of tissues
C) Atoms form molecules through chemical bonding ✓
D) Atoms exist only in plants
6. Which statement best describes molecules in cells?
A) Molecules are made from bonded atoms ✓
B) Molecules cannot participate in reactions
C) Molecules are larger than organ systems
D) Molecules contain no energy
7. Which statement correctly explains the importance of carbon in cells?
A) Carbon cannot form stable bonds
B) Carbon forms the backbone of many biological molecules ✓
C) Carbon exists only in inorganic matter
D) Carbon has no role in metabolism

8. Which statement about water in cells is correct?

- A) Water has no biological role
- B) Water acts as an important solvent in cells ✓
- C) Cells contain very little water
- D) Water prevents cellular reactions

9. Which statement best explains the journey into the cell using microscopes?

- A) Microscopes made it possible to observe cell structures ✓
- B) Microscopes destroy cellular components
- C) Cells can only be studied without instruments
- D) Microscopes are used only for plants

10. Which statement about electron microscopes is correct?

- A) They provide lower magnification than light microscopes
- B) They cannot observe organelles
- C) They provide highly detailed images of cell structures ✓
- D) They use sound waves for imaging

11. Which statement correctly compares prokaryotic and eukaryotic cells?

- A) Prokaryotes possess membrane-bound nuclei
- B) Eukaryotic cells contain membrane-bound organelles ✓
- C) Both lack genetic material
- D) Prokaryotes are always larger than eukaryotes

12. Which statement best explains why cells remain small in size?

- A) Small cells maintain efficient exchange of materials ✓
- B) Large cells cannot contain DNA
- C) Small cells have no metabolism
- D) Cell size has no biological importance

13. Which statement correctly describes the surface area-to-volume ratio in cells?

- A) Larger cells have a higher surface area-to-volume ratio
- B) Smaller cells exchange substances more efficiently ✓
- C) Surface area decreases metabolism completely
- D) Volume has no effect on cell survival

14. Which statement about cell size limitation is correct?

- A) Diffusion becomes inefficient in excessively large cells ✓
- B) Large cells always function better
- C) Cell size does not affect transport
- D) Diffusion increases with unlimited growth

15. Which statement best describes the importance of cells in multicellular organisms?

- A) Cells work independently without coordination
- B) Specialized cells perform specific functions ✓
- C) All cells have identical roles
- D) Cells are unnecessary for tissues

16. Which statement about organelles is correct?

- A) Organelles perform specialized tasks within cells ✓

- B) Organelles exist outside the cell membrane
 - C) Organelles are found only in bacteria
 - D) Organelles have no metabolic role
17. Which statement correctly explains why the nucleus is important?
- A) It produces cellular waste only
 - B) It stores and controls genetic information ✓
 - C) It is absent in all eukaryotes
 - D) It performs photosynthesis
18. Which statement about the plasma membrane is correct?
- A) It allows unrestricted movement of all substances
 - B) It regulates movement into and out of the cell ✓
 - C) It exists only in plant cells
 - D) It contains no proteins
19. Which statement best explains the importance of studying cells at university level?
- A) Cell biology helps understand health and disease ✓
 - B) Cells are unrelated to medicine
 - C) Cellular studies have no practical applications
 - D) Only botanists study cells
20. Which statement about modern cell biology is correct?
- A) Cell biology contributes to biotechnology and medicine ✓
 - B) Cell biology has no role in research
 - C) Cells cannot be genetically studied
 - D) Modern science ignores cellular processes
51. Which statement correctly describes the fluid mosaic model of the cell membrane?
- A) The membrane is rigid and immovable
 - B) The membrane contains moving lipids and proteins ✓
 - C) The membrane is made entirely of carbohydrates
 - D) The membrane has no organization
52. Which statement about phospholipids is correct?
- A) Phospholipids form the basic structure of cell membranes ✓
 - B) Phospholipids contain only proteins
 - C) Phospholipids are insoluble minerals
 - D) Phospholipids prevent membrane flexibility
53. Which statement best explains facilitated diffusion?
- A) It requires ATP for transport
 - B) It uses transport proteins to move substances passively ✓
 - C) It occurs only in dead cells
 - D) It transports substances against concentration gradient
54. Which statement correctly describes endocytosis?
- A) Endocytosis allows cells to engulf materials ✓
 - B) Endocytosis releases wastes only

- C) Endocytosis occurs without membranes
D) Endocytosis destroys ATP production
55. Which statement about exocytosis is correct?
A) It exports substances out of the cell ✓
B) It imports nutrients only
C) It occurs only in bacteria
D) It stops secretion completely
56. Which statement best explains cellular respiration?
A) Cells convert nutrients into usable energy ✓
B) Cells produce oxygen during respiration
C) Respiration occurs only in plants
D) Respiration prevents ATP formation
57. Which statement correctly describes aerobic respiration?
A) It requires oxygen for ATP production ✓
B) It occurs without oxygen only
C) It produces no energy
D) It occurs outside the cell
58. Which statement about anaerobic respiration is correct?
A) It occurs in the absence of oxygen ✓
B) It generates more ATP than aerobic respiration always
C) It only occurs in plants
D) It stops glucose breakdown
59. Which statement best explains photosynthesis?
A) Light energy is converted into chemical energy ✓
B) Photosynthesis destroys glucose
C) Photosynthesis occurs only in animals
D) Photosynthesis requires no chlorophyll
60. Which statement correctly describes chlorophyll?
A) Chlorophyll absorbs light energy for photosynthesis ✓
B) Chlorophyll produces ATP directly in animals
C) Chlorophyll is a structural protein
D) Chlorophyll prevents plant growth
61. Which statement about cell communication is correct?
A) Cells communicate using chemical signals ✓
B) Cells function without coordination
C) Cells cannot respond to hormones
D) Communication occurs only in bacteria
62. Which statement best explains receptors in cells?
A) Receptors detect specific signals ✓
B) Receptors synthesize DNA
C) Receptors destroy hormones
D) Receptors prevent cell interaction

63. Which statement correctly describes mitosis?
- A) Mitosis produces genetically identical cells ✓
 - B) Mitosis produces gametes only
 - C) Mitosis halves chromosome number
 - D) Mitosis occurs only in prokaryotes
64. Which statement about meiosis is correct?
- A) Meiosis produces cells with half chromosome number ✓
 - B) Meiosis produces identical body cells
 - C) Meiosis occurs in bacteria only
 - D) Meiosis prevents genetic variation
65. Which statement best explains the importance of chromosomes?
- A) Chromosomes carry genetic information ✓
 - B) Chromosomes produce ATP only
 - C) Chromosomes are made entirely of lipids
 - D) Chromosomes exist outside nuclei only
66. Which statement correctly describes genes?
- A) Genes are units of hereditary information ✓
 - B) Genes are membrane proteins
 - C) Genes contain no DNA
 - D) Genes exist only in plants
67. Which statement about mutations is correct?
- A) Mutations are changes in genetic material ✓
 - B) Mutations always improve organisms
 - C) Mutations occur only in animals
 - D) Mutations cannot affect proteins
68. Which statement best explains apoptosis?
- A) Apoptosis is programmed cell death ✓
 - B) Apoptosis causes uncontrolled cell growth
 - C) Apoptosis occurs only in bacteria
 - D) Apoptosis prevents development completely
69. Which statement correctly describes cancer cells?
- A) Cancer cells divide uncontrollably ✓
 - B) Cancer cells never reproduce
 - C) Cancer cells lack DNA
 - D) Cancer cells perform only photosynthesis
70. Which statement about stem cell therapy is correct?
- A) Stem cells may help repair damaged tissues ✓
 - B) Stem cells cannot divide
 - C) Stem cell therapy has no medical applications
 - D) Stem cells exist only in embryos
71. Which statement best explains tissue formation?
- A) Similar cells combine to perform specific functions ✓

- B) Tissues are made of atoms only
- C) Tissues cannot form organs
- D) Tissues contain no living cells

72. Which statement correctly describes organs?

- A) Organs consist of different tissues working together ✓
- B) Organs are made of one cell only
- C) Organs function independently without tissues
- D) Organs contain no blood supply

73. Which statement about organ systems is correct?

- A) Organ systems coordinate body functions ✓
- B) Organ systems exist only in plants
- C) Organ systems contain no organs
- D) Organ systems prevent homeostasis

74. Which statement best explains the hierarchy of biological organization?

- A) Cells form tissues, tissues form organs ✓
- B) Organs form atoms directly
- C) Molecules are larger than organs
- D) Tissues form atoms

75. Which statement correctly describes cytoskeleton function?

- A) It provides shape and support to cells ✓
- B) It performs photosynthesis only
- C) It prevents intracellular movement
- D) It contains chlorophyll

76. Which statement about microtubules is correct?

- A) They help in intracellular transport and cell division ✓
- B) They produce glucose only
- C) They are found outside cells
- D) They contain no proteins

77. Which statement best explains cilia and flagella?

- A) They assist in cellular movement ✓
- B) They store genetic information
- C) They prevent transport
- D) They exist only in plant roots

78. Which statement correctly describes bacterial cells?

- A) Bacteria are prokaryotic organisms ✓
- B) Bacteria possess membrane-bound nuclei
- C) Bacteria lack ribosomes
- D) Bacteria are multicellular only

79. Which statement about viruses is correct?

- A) Viruses require host cells for replication ✓
- B) Viruses perform independent metabolism

- C) Viruses are made of tissues
- D) Viruses contain organ systems

80. Which statement best explains the role of ATP in metabolism?

- A) ATP transfers usable energy within cells ✓
- B) ATP stores hereditary information
- C) ATP blocks enzyme activity
- D) ATP forms cell walls

81. Which statement correctly describes enzymes as catalysts?

- A) Enzymes lower activation energy of reactions ✓
- B) Enzymes stop all reactions
- C) Enzymes are consumed permanently
- D) Enzymes are inorganic salts only

82. Which statement about pH and enzymes is correct?

- A) Enzymes function best within specific pH ranges ✓
- B) pH never affects enzymes
- C) All enzymes work only in acidic conditions
- D) Enzymes function without proteins

83. Which statement best explains denaturation of proteins?

- A) Protein structure is altered by extreme conditions ✓
- B) Proteins become stronger permanently
- C) Denaturation increases enzyme efficiency always
- D) Denaturation occurs only in plants

84. Which statement correctly describes passive transport?

- A) Passive transport requires no cellular energy ✓
- B) Passive transport always requires ATP
- C) Passive transport occurs against gradients only
- D) Passive transport stops diffusion

85. Which statement about isotonic solutions is correct?

- A) Water movement is balanced across the membrane ✓
- B) Cells always burst in isotonic solution
- C) Isotonic solutions contain no water
- D) Isotonic conditions stop osmosis completely

86. Which statement best explains hypotonic solutions?

- A) Water enters the cell causing swelling ✓
- B) Water leaves the cell completely
- C) Hypotonic solutions contain no solutes
- D) Hypotonic conditions shrink cells always

87. Which statement correctly describes hypertonic solutions?

- A) Water leaves the cell causing shrinkage ✓
- B) Cells gain unlimited water
- C) Hypertonic solutions prevent osmosis
- D) Hypertonic solutions contain pure water only

88. Which statement about cell specialization is correct?

- A) Specialized cells perform unique biological functions ✓
- B) All cells perform identical tasks
- C) Specialization prevents growth
- D) Specialization removes DNA

89. Which statement best explains nerve cell adaptation?

- A) Neurons are specialized for rapid signal transmission ✓
- B) Neurons perform photosynthesis
- C) Neurons lack membranes
- D) Neurons store chlorophyll

90. Which statement correctly describes muscle cells?

- A) Muscle cells are specialized for contraction ✓
- B) Muscle cells cannot use ATP
- C) Muscle cells lack proteins
- D) Muscle cells perform only digestion

91. Which statement about red blood cells is correct?

- A) They transport oxygen throughout the body ✓
- B) They perform photosynthesis
- C) They contain chloroplasts
- D) They produce hormones only

92. Which statement best explains immune cells?

- A) Immune cells defend the body against pathogens ✓
- B) Immune cells produce plant pigments
- C) Immune cells prevent circulation
- D) Immune cells contain no nucleus

93. Which statement correctly describes extracellular matrix?

- A) It provides support outside cells ✓
- B) It exists only inside nuclei
- C) It prevents tissue organization
- D) It is absent in animals

94. Which statement about biotechnology is correct?

- A) Biotechnology applies cellular knowledge for practical uses ✓
- B) Biotechnology has no medical role
- C) Biotechnology ignores genetics
- D) Biotechnology studies only fossils

95. Which statement best explains genetic engineering?

- A) Genes are modified for desired purposes ✓
- B) Genetic engineering destroys all DNA
- C) Genetic engineering occurs naturally only
- D) Genetic engineering prevents protein synthesis

96. Which statement correctly describes cloning?

- A) Cloning produces genetically identical copies ✓

- B) Cloning always creates mutations
- C) Cloning occurs only in plants
- D) Cloning removes chromosomes

97. Which statement about microscopy is correct?

- A) Advanced microscopes improved understanding of cells ✓
- B) Microscopy has no role in biology
- C) Microscopes cannot magnify organelles
- D) Microscopy studies only chemicals

98. Which statement best explains why cells are called building blocks of life?

- A) All living organisms are composed of cells ✓
- B) Cells exist only in animals
- C) Cells cannot reproduce
- D) Cells are non-living structures

99. Which statement correctly describes the relationship between structure and function in cells?

- A) Cell structure is closely related to its function ✓
- B) Structure has no effect on function
- C) All cells have identical structures
- D) Function changes without structure

100. Which statement about modern cell biology is correct?

- A) It integrates genetics, biochemistry, and molecular biology ✓
- B) It studies only visible organisms
- C) It ignores molecular processes
- D) It has no role in scientific advancement

1. Which statement best defines biomolecules?

- A) Biomolecules are non-living minerals only
- B) Biomolecules are chemical compounds found in living organisms ✓
- C) Biomolecules exist only in rocks
- D) Biomolecules cannot contain carbon

2. Which statement correctly describes proteins?

- A) Proteins are polymers of amino acids ✓
- B) Proteins are made entirely of glucose
- C) Proteins contain no nitrogen
- D) Proteins are inorganic salts

3. Which statement about amino acids is correct?

- A) Amino acids are building blocks of proteins ✓
- B) Amino acids are simple sugars
- C) Amino acids cannot form bonds
- D) Amino acids exist only in plants

4. Which statement correctly explains peptide bonds?

- A) Peptide bonds link amino acids together ✓
- B) Peptide bonds connect lipids only

- C) Peptide bonds prevent protein formation
D) Peptide bonds contain no nitrogen
5. Which statement best explains the primary structure of proteins?
A) It is the sequence of amino acids in a polypeptide ✓
B) It refers to protein folding only
C) It is the final 3D shape of proteins
D) It involves multiple polypeptide chains only
6. Which statement about secondary protein structure is correct?
A) It includes alpha-helices and beta-sheets ✓
B) It is formed by DNA replication
C) It exists only in enzymes
D) It prevents hydrogen bonding
7. Which statement correctly describes tertiary protein structure?
A) It is the three-dimensional folding of a polypeptide ✓
B) It contains no side chain interactions
C) It is limited to amino acid sequence only
D) It prevents biological activity
8. Which statement about quaternary protein structure is correct?
A) It involves interaction of multiple polypeptide chains ✓
B) It occurs only in lipids
C) It contains no subunits
D) It is simpler than primary structure
9. Which statement best explains denaturation of proteins?
A) Denaturation alters protein shape and function ✓
B) Denaturation strengthens proteins permanently
C) Denaturation forms peptide bonds
D) Denaturation increases all enzyme activity
10. Which statement correctly describes enzymes?
A) Enzymes are proteins that catalyze reactions ✓
B) Enzymes stop metabolism completely
C) Enzymes are carbohydrates only
D) Enzymes cannot bind substrates
11. Which statement about structural proteins is correct?
A) Structural proteins provide support and strength ✓
B) Structural proteins transport oxygen only
C) Structural proteins contain no amino acids
D) Structural proteins cannot form fibers
12. Which statement correctly describes collagen?
A) Collagen is a structural protein in connective tissues ✓
B) Collagen stores genetic information
C) Collagen is a membrane lipid
D) Collagen performs photosynthesis

13. Which statement about keratin is correct?
A) Keratin is found in hair, nails, and skin ✓
B) Keratin transports glucose
C) Keratin is a carbohydrate
D) Keratin exists only in bacteria
14. Which statement best explains the importance of proteins in growth?
A) Proteins are essential for tissue development and repair ✓
B) Proteins prevent cell formation
C) Proteins contain no energy
D) Proteins stop metabolism
15. Which statement correctly describes transport proteins?
A) Transport proteins move substances within organisms ✓
B) Transport proteins produce ATP only
C) Transport proteins are lipids
D) Transport proteins destroy nutrients
16. Which statement about hemoglobin is correct?
A) Hemoglobin transports oxygen in blood ✓
B) Hemoglobin performs digestion
C) Hemoglobin contains glucose only
D) Hemoglobin prevents respiration
17. Which statement correctly explains antibodies?
A) Antibodies are proteins involved in immunity ✓
B) Antibodies synthesize DNA
C) Antibodies store fats
D) Antibodies prevent immune response
18. Which statement about contractile proteins is correct?
A) Contractile proteins help muscles contract ✓
B) Contractile proteins perform photosynthesis
C) Contractile proteins exist only in plants
D) Contractile proteins contain no amino acids
19. Which statement correctly describes actin and myosin?
A) They are muscle proteins involved in movement ✓
B) They are carbohydrate molecules
C) They transport oxygen only
D) They prevent contraction
20. Which statement best explains the role of proteins in metabolism?
A) Proteins regulate and support biochemical reactions ✓
B) Proteins stop cellular respiration
C) Proteins destroy enzymes
D) Proteins prevent ATP production
21. Which statement correctly describes globular proteins?
A) Globular proteins are compact and soluble ✓

- B) Globular proteins are always fibrous
 - C) Globular proteins cannot function as enzymes
 - D) Globular proteins contain no tertiary structure
22. Which statement about fibrous proteins is correct?
- A) Fibrous proteins mainly provide structural support ✓
 - B) Fibrous proteins dissolve easily in water
 - C) Fibrous proteins transport hormones only
 - D) Fibrous proteins contain no peptide bonds
23. Which statement correctly explains protein synthesis?
- A) Proteins are synthesized using genetic information ✓
 - B) Proteins form without ribosomes
 - C) Protein synthesis occurs outside cells only
 - D) Protein synthesis destroys amino acids
24. Which statement about ribosomes is correct?
- A) Ribosomes are sites of protein synthesis ✓
 - B) Ribosomes produce lipids only
 - C) Ribosomes lack RNA
 - D) Ribosomes prevent translation
25. Which statement correctly describes essential amino acids?
- A) Essential amino acids must be obtained from diet ✓
 - B) Essential amino acids are produced in sufficient amounts by the body
 - C) Essential amino acids contain no nitrogen
 - D) Essential amino acids exist only in plants
26. Which statement about non-essential amino acids is correct?
- A) They can be synthesized by the body ✓
 - B) They cannot participate in proteins
 - C) They are absent in humans
 - D) They only exist in food
27. Which statement correctly explains complete proteins?
- A) Complete proteins contain all essential amino acids ✓
 - B) Complete proteins contain carbohydrates only
 - C) Complete proteins lack nitrogen
 - D) Complete proteins are inorganic compounds
28. Which statement about incomplete proteins is correct?
- A) They lack one or more essential amino acids ✓
 - B) They contain no peptide bonds
 - C) They cannot be digested
 - D) They are toxic compounds
29. Which statement best explains protein digestion?
- A) Proteins are broken into amino acids by enzymes ✓
 - B) Proteins remain unchanged during digestion

- C) Proteins are converted directly into DNA
- D) Proteins bypass the digestive system

30. Which statement correctly describes proteases?

- A) Proteases are enzymes that digest proteins ✓
- B) Proteases synthesize carbohydrates
- C) Proteases transport oxygen
- D) Proteases are lipids only

31. Which statement about denatured proteins is correct?

- A) They may lose biological function ✓
- B) They gain unlimited activity
- C) They form DNA automatically
- D) They become carbohydrates

32. Which statement correctly explains the role of proteins in cell signaling?

- A) Proteins help transmit cellular signals ✓
- B) Proteins prevent communication
- C) Proteins stop hormone activity
- D) Proteins destroy receptors

33. Which statement about receptor proteins is correct?

- A) Receptor proteins bind specific molecules ✓
- B) Receptor proteins are inorganic salts
- C) Receptor proteins prevent transport
- D) Receptor proteins contain no amino acids

34. Which statement correctly describes hormonal proteins?

- A) Some hormones are proteins regulating body functions ✓
- B) All hormones are lipids only
- C) Protein hormones cannot affect cells
- D) Hormonal proteins lack specificity

35. Which statement best explains insulin?

- A) Insulin is a protein hormone regulating blood glucose ✓
- B) Insulin stores oxygen
- C) Insulin is a carbohydrate enzyme
- D) Insulin prevents metabolism

36. Which statement about protein structure and function is correct?

- A) Protein function depends on its shape ✓
- B) Protein shape has no effect on function
- C) All proteins have identical shapes
- D) Protein function depends only on water

37. Which statement correctly explains hydrogen bonds in proteins?

- A) Hydrogen bonds stabilize protein structures ✓
- B) Hydrogen bonds destroy amino acids
- C) Hydrogen bonds exist only in lipids
- D) Hydrogen bonds prevent folding

38. Which statement about disulfide bonds is correct?
A) They help stabilize tertiary protein structure ✓
B) They occur only in carbohydrates
C) They prevent protein folding
D) They contain no sulfur
39. Which statement correctly describes enzyme specificity?
A) Enzymes act on specific substrates ✓
B) Enzymes react with all molecules equally
C) Enzymes prevent metabolism
D) Enzymes function without active sites
40. Which statement best explains the lock-and-key model?
A) Substrates fit precisely into enzyme active sites ✓
B) Enzymes permanently bind all molecules
C) Active sites contain DNA only
D) Enzymes stop reactions permanently
41. Which statement about the induced-fit model is correct?
A) Enzymes adjust shape slightly to bind substrates ✓
B) Enzymes remain completely rigid
C) Enzymes cannot bind substrates
D) Induced-fit prevents catalysis
42. Which statement correctly explains protein malnutrition?
A) Protein deficiency can impair growth and repair ✓
B) Protein deficiency increases muscle mass
C) Protein deficiency has no health effects
D) Protein deficiency improves immunity
43. Which statement about kwashiorkor is correct?
A) Kwashiorkor results from severe protein deficiency ✓
B) Kwashiorkor is caused by excess vitamins
C) Kwashiorkor improves muscle development
D) Kwashiorkor is unrelated to nutrition
44. Which statement correctly describes marasmus?
A) Marasmus results from severe calorie and protein deficiency ✓
B) Marasmus increases body fat
C) Marasmus occurs due to excess proteins
D) Marasmus prevents weight loss
45. Which statement best explains the buffering role of proteins?
A) Proteins help maintain pH balance ✓
B) Proteins prevent water movement
C) Proteins destroy ions
D) Proteins stop respiration
46. Which statement about storage proteins is correct?
A) Storage proteins store amino acids or nutrients ✓

- B) Storage proteins transport oxygen only
- C) Storage proteins lack peptide bonds
- D) Storage proteins prevent growth

47. Which statement correctly describes casein?

- A) Casein is a storage protein in milk ✓
- B) Casein is a structural carbohydrate
- C) Casein transports glucose only
- D) Casein performs photosynthesis

48. Which statement about albumin is correct?

- A) Albumin helps maintain blood osmotic balance ✓
- B) Albumin is a DNA molecule
- C) Albumin prevents circulation
- D) Albumin contains no amino acids

49. Which statement correctly explains protein turnover?

- A) Proteins are continuously synthesized and degraded ✓
- B) Proteins remain unchanged forever
- C) Protein turnover stops metabolism
- D) Protein turnover occurs only in plants

50. Which statement about protein advantages is correct?

- A) Proteins support growth, repair, and metabolism ✓
- B) Proteins have no biological importance
- C) Proteins cannot provide structure
- D) Proteins stop enzyme activity

51. Which statement correctly describes catalytic proteins?

- A) Catalytic proteins accelerate biochemical reactions ✓
- B) Catalytic proteins store genetic information
- C) Catalytic proteins are carbohydrates only
- D) Catalytic proteins stop metabolism

52. Which statement about amino acid side chains is correct?

- A) Side chains determine amino acid properties ✓
- B) Side chains are identical in all amino acids
- C) Side chains contain no atoms
- D) Side chains prevent peptide bond formation

53. Which statement correctly explains hydrophobic interactions in proteins?

- A) Hydrophobic regions tend to avoid water during folding ✓
- B) Hydrophobic interactions occur only in DNA
- C) Hydrophobic groups dissolve easily in water
- D) Hydrophobic interactions prevent tertiary structure

54. Which statement about hydrophilic amino acids is correct?

- A) They interact favorably with water ✓
- B) They repel all liquids

- C) They contain no charges
- D) They prevent protein solubility

55. Which statement correctly describes conjugated proteins?

- A) Conjugated proteins contain a non-protein component ✓
- B) Conjugated proteins lack amino acids
- C) Conjugated proteins are carbohydrates only
- D) Conjugated proteins cannot function biologically

56. Which statement about simple proteins is correct?

- A) Simple proteins yield only amino acids upon hydrolysis ✓
- B) Simple proteins contain no peptide bonds
- C) Simple proteins are inorganic compounds
- D) Simple proteins prevent digestion

57. Which statement correctly explains glycoproteins?

- A) Glycoproteins contain carbohydrate groups attached to proteins ✓
- B) Glycoproteins contain no amino acids
- C) Glycoproteins exist only in plants
- D) Glycoproteins stop cellular communication

58. Which statement about lipoproteins is correct?

- A) Lipoproteins transport lipids in blood ✓
- B) Lipoproteins synthesize DNA
- C) Lipoproteins contain no proteins
- D) Lipoproteins prevent metabolism

59. Which statement correctly describes chromoproteins?

- A) Chromoproteins contain colored prosthetic groups ✓
- B) Chromoproteins are colorless carbohydrates
- C) Chromoproteins cannot transport oxygen
- D) Chromoproteins lack tertiary structure

60. Which statement about metalloproteins is correct?

- A) Metalloproteins contain metal ions as part of structure ✓
- B) Metalloproteins contain no amino acids
- C) Metalloproteins exist only in bacteria
- D) Metalloproteins prevent enzyme activity

61. Which statement correctly explains enzyme inhibition?

- A) Inhibitors reduce enzyme activity ✓
- B) Inhibitors permanently increase all reactions
- C) Inhibitors are always proteins
- D) Inhibitors prevent substrate binding only through destruction

62. Which statement about competitive inhibition is correct?

- A) Competitive inhibitors compete with substrate for active site ✓
- B) Competitive inhibitors destroy enzymes permanently
- C) Competitive inhibitors bind DNA only
- D) Competitive inhibition increases substrate binding

63. Which statement correctly describes non-competitive inhibition?

- A) Inhibitors bind away from active site altering enzyme function ✓
- B) Inhibitors occupy substrate position directly
- C) Non-competitive inhibitors strengthen catalysis
- D) Non-competitive inhibition occurs only in plants

64. Which statement about allosteric proteins is correct?

- A) Their activity changes when molecules bind regulatory sites ✓
- B) They cannot change shape
- C) They contain no amino acids
- D) They are inactive permanently

65. Which statement correctly explains protein folding?

- A) Protein folding determines functional three-dimensional shape ✓
- B) Folding destroys peptide bonds
- C) Folding prevents biological activity
- D) Folding occurs only outside cells

66. Which statement about chaperone proteins is correct?

- A) Chaperones assist proper protein folding ✓
- B) Chaperones digest proteins completely
- C) Chaperones prevent all molecular interactions
- D) Chaperones are carbohydrates only

67. Which statement correctly describes irreversible denaturation?

- A) Protein cannot regain original structure and function ✓
- B) Protein becomes more active permanently
- C) Protein transforms into DNA
- D) Protein synthesis stops forever

68. Which statement about reversible denaturation is correct?

- A) Some proteins can regain original structure under proper conditions ✓
- B) Reversible denaturation destroys amino acids permanently
- C) Reversible denaturation prevents folding forever
- D) Reversible denaturation occurs only in plants

69. Which statement correctly explains the nutritional importance of proteins?

- A) Proteins are essential for body maintenance and repair ✓
- B) Proteins provide no biological benefits
- C) Proteins cannot supply energy
- D) Proteins are unnecessary in diet

70. Which statement about protein-rich foods is correct?

- A) Meat, eggs, beans, and milk are rich protein sources ✓
- B) Proteins are found only in fruits
- C) Protein foods contain no amino acids
- D) Protein foods cannot support growth

71. Which statement correctly describes soybean protein?

- A) Soybean is a major plant source of protein ✓

- B) Soybean contains no amino acids
- C) Soybean protein lacks nutritional value
- D) Soybean is composed entirely of fats

72. Which statement about animal proteins is correct?

- A) Animal proteins are usually complete proteins ✓
- B) Animal proteins contain no nitrogen
- C) Animal proteins prevent growth
- D) Animal proteins are indigestible

73. Which statement correctly explains plant proteins?

- A) Many plant proteins are incomplete proteins ✓
- B) Plant proteins contain no amino acids
- C) Plant proteins cannot support metabolism
- D) Plant proteins are always toxic

74. Which statement about nitrogen balance is correct?

- A) Nitrogen balance reflects protein metabolism in the body ✓
- B) Nitrogen balance measures lipid digestion only
- C) Nitrogen balance has no nutritional importance
- D) Nitrogen balance prevents amino acid synthesis

75. Which statement correctly explains positive nitrogen balance?

- A) Nitrogen intake exceeds nitrogen loss ✓
- B) Nitrogen loss exceeds intake
- C) Protein synthesis stops completely
- D) Positive balance prevents growth

76. Which statement about negative nitrogen balance is correct?

- A) Nitrogen loss exceeds intake ✓
- B) Nitrogen intake exceeds excretion
- C) Negative balance improves tissue growth
- D) Negative balance increases protein stores

77. Which statement correctly describes protein hydrolysis?

- A) Protein hydrolysis breaks peptide bonds using water ✓
- B) Protein hydrolysis forms DNA
- C) Protein hydrolysis prevents digestion
- D) Protein hydrolysis occurs without enzymes

78. Which statement about amino acid metabolism is correct?

- A) Amino acids can be used for energy production ✓
- B) Amino acids cannot participate in metabolism
- C) Amino acids are stored permanently without use
- D) Amino acids exist only in proteins

79. Which statement correctly explains transamination?

- A) Amino groups are transferred between molecules ✓
- B) Transamination destroys amino acids completely

- C) Transamination forms carbohydrates only
- D) Transamination occurs only in plants

80. Which statement about deamination is correct?

- A) Deamination removes amino groups from amino acids ✓
- B) Deamination synthesizes proteins directly
- C) Deamination prevents energy production
- D) Deamination occurs only outside cells

81. Which statement correctly describes urea formation?

- A) Urea is formed from ammonia during amino acid breakdown ✓
- B) Urea stores oxygen in blood
- C) Urea prevents excretion
- D) Urea is a structural protein

82. Which statement about plasma proteins is correct?

- A) Plasma proteins help maintain osmotic pressure ✓
- B) Plasma proteins contain no amino acids
- C) Plasma proteins stop blood circulation
- D) Plasma proteins prevent clotting

83. Which statement correctly explains fibrinogen?

- A) Fibrinogen is involved in blood clotting ✓
- B) Fibrinogen transports oxygen only
- C) Fibrinogen is a carbohydrate
- D) Fibrinogen prevents wound healing

84. Which statement about membrane proteins is correct?

- A) Membrane proteins assist transport and communication ✓
- B) Membrane proteins exist only outside cells
- C) Membrane proteins prevent diffusion completely
- D) Membrane proteins contain no peptide bonds

85. Which statement correctly describes channel proteins?

- A) Channel proteins allow passage of specific ions or molecules ✓
- B) Channel proteins digest proteins only
- C) Channel proteins block all transport
- D) Channel proteins are lipids only

86. Which statement about carrier proteins is correct?

- A) Carrier proteins transport substances across membranes ✓
- B) Carrier proteins synthesize ATP directly
- C) Carrier proteins contain no amino acids
- D) Carrier proteins stop osmosis

87. Which statement correctly explains motor proteins?

- A) Motor proteins generate movement within cells ✓
- B) Motor proteins prevent muscle contraction
- C) Motor proteins are carbohydrates
- D) Motor proteins exist only in bacteria

88. Which statement about kinesin is correct?

- A) Kinesin transports materials along microtubules ✓
- B) Kinesin stores glucose
- C) Kinesin performs photosynthesis
- D) Kinesin prevents intracellular transport

89. Which statement correctly describes dynein?

- A) Dynein is involved in movement of cilia and flagella ✓
- B) Dynein synthesizes proteins only
- C) Dynein prevents cellular motion
- D) Dynein is a lipid molecule

90. Which statement about elastin is correct?

- A) Elastin provides elasticity to tissues ✓
- B) Elastin transports oxygen
- C) Elastin is a carbohydrate polymer
- D) Elastin prevents flexibility

91. Which statement correctly explains protein advantages in sports?

- A) Proteins help repair and build muscle tissues ✓
- B) Proteins prevent energy production
- C) Proteins weaken muscles permanently
- D) Proteins stop tissue recovery

92. Which statement about whey protein is correct?

- A) Whey protein is commonly used as a dietary supplement ✓
- B) Whey protein contains no amino acids
- C) Whey protein prevents growth
- D) Whey protein is an inorganic compound

93. Which statement correctly describes protein supplements?

- A) Protein supplements may help meet dietary protein needs ✓
- B) Protein supplements contain no nutrients
- C) Protein supplements stop metabolism
- D) Protein supplements replace water completely

94. Which statement about enzymes and temperature is correct?

- A) High temperatures may denature enzymes ✓
- B) Temperature never affects enzymes
- C) Enzymes work best at all temperatures equally
- D) Low temperatures permanently destroy all enzymes

95. Which statement correctly explains optimum pH?

- A) Each enzyme functions best at a specific pH ✓
- B) All enzymes work only at pH 1
- C) pH has no effect on catalysis
- D) Optimum pH destroys enzymes

96. Which statement about active sites is correct?

- A) Active sites are regions where substrates bind ✓

- B) Active sites contain DNA only
- C) Active sites prevent catalysis
- D) Active sites exist outside enzymes only

97. Which statement correctly describes substrate concentration effect on enzymes?

- A) Increasing substrate concentration raises reaction rate until saturation ✓
- B) Substrate concentration never affects enzymes
- C) Excess substrate permanently denatures enzymes
- D) Reaction rate decreases immediately with all substrate increases

98. Which statement about cofactors is correct?

- A) Cofactors assist enzyme activity ✓
- B) Cofactors prevent catalysis
- C) Cofactors are always proteins
- D) Cofactors destroy substrates

99. Which statement correctly explains coenzymes?

- A) Coenzymes are organic molecules assisting enzymes ✓
- B) Coenzymes are structural carbohydrates
- C) Coenzymes permanently inhibit enzymes
- D) Coenzymes contain no carbon

100. Which statement about proteins and life processes is correct?

- A) Proteins are essential for nearly all cellular functions ✓
- B) Proteins have no role in metabolism
- C) Proteins are unnecessary for survival
- D) Proteins exist only in muscles

1. Which statement best defines carbohydrates?

- A) Carbohydrates are inorganic salts
- B) Carbohydrates are organic compounds containing carbon, hydrogen, and oxygen ✓
- C) Carbohydrates contain nitrogen only
- D) Carbohydrates are proteins only

2. Which statement correctly describes the main function of carbohydrates?

- A) Carbohydrates mainly provide energy ✓
- B) Carbohydrates store genetic information
- C) Carbohydrates prevent metabolism
- D) Carbohydrates contain no calories

3. Which statement about monosaccharides is correct?

- A) Monosaccharides are the simplest carbohydrates ✓
- B) Monosaccharides contain many sugar units
- C) Monosaccharides cannot dissolve in water
- D) Monosaccharides are proteins

4. Which statement correctly describes glucose?

- A) Glucose is a major energy source for cells ✓
- B) Glucose is a lipid molecule

- C) Glucose prevents respiration
D) Glucose contains amino acids
5. Which statement about fructose is correct?
A) Fructose is commonly found in fruits ✓
B) Fructose is a protein
C) Fructose contains nitrogen only
D) Fructose cannot provide energy
6. Which statement correctly describes galactose?
A) Galactose is a monosaccharide found in milk sugar ✓
B) Galactose is a polysaccharide
C) Galactose cannot combine with glucose
D) Galactose prevents digestion
7. Which statement about disaccharides is correct?
A) Disaccharides consist of two monosaccharides joined together ✓
B) Disaccharides contain no sugar units
C) Disaccharides are proteins only
D) Disaccharides cannot be digested
8. Which statement correctly describes sucrose?
A) Sucrose is composed of glucose and fructose ✓
B) Sucrose contains amino acids only
C) Sucrose is an enzyme
D) Sucrose is insoluble in water
9. Which statement about lactose is correct?
A) Lactose is the sugar found in milk ✓
B) Lactose is composed of proteins
C) Lactose contains no energy
D) Lactose prevents digestion
10. Which statement correctly describes maltose?
A) Maltose consists of two glucose molecules ✓
B) Maltose is a lipid
C) Maltose cannot be hydrolyzed
D) Maltose stores oxygen
11. Which statement about polysaccharides is correct?
A) Polysaccharides contain many monosaccharide units ✓
B) Polysaccharides are simple sugars only
C) Polysaccharides cannot store energy
D) Polysaccharides contain amino acids
12. Which statement correctly describes starch?
A) Starch is the storage carbohydrate in plants ✓
B) Starch is found only in animals
C) Starch is a protein polymer
D) Starch cannot be digested

13. Which statement about glycogen is correct?
- A) Glycogen is the storage carbohydrate in animals ✓
 - B) Glycogen exists only in plants
 - C) Glycogen contains no glucose
 - D) Glycogen prevents energy release
14. Which statement correctly describes cellulose?
- A) Cellulose is a structural polysaccharide in plant cell walls ✓
 - B) Cellulose stores energy in animals
 - C) Cellulose is a protein
 - D) Cellulose contains amino acids
15. Which statement about chitin is correct?
- A) Chitin is found in fungal cell walls and arthropod exoskeletons ✓
 - B) Chitin is a lipid molecule
 - C) Chitin stores oxygen
 - D) Chitin prevents structural support
16. Which statement correctly explains carbohydrate digestion?
- A) Carbohydrates are broken down into simple sugars ✓
 - B) Carbohydrates remain unchanged during digestion
 - C) Carbohydrates are converted directly into proteins
 - D) Carbohydrate digestion occurs only in plants
17. Which statement about amylase is correct?
- A) Amylase is an enzyme that digests starch ✓
 - B) Amylase digests proteins only
 - C) Amylase prevents glucose formation
 - D) Amylase is a lipid
18. Which statement correctly explains carbohydrate metabolism?
- A) Carbohydrates are broken down to release energy ✓
 - B) Carbohydrates cannot participate in respiration
 - C) Carbohydrates stop ATP production
 - D) Carbohydrates are not metabolized
19. Which statement about ATP production is correct?
- A) Glucose metabolism produces ATP ✓
 - B) ATP prevents energy release
 - C) ATP is a carbohydrate polymer
 - D) ATP contains no energy
20. Which statement correctly describes glycolysis?
- A) Glycolysis is the breakdown of glucose for energy ✓
 - B) Glycolysis produces proteins
 - C) Glycolysis occurs only in plants
 - D) Glycolysis prevents ATP synthesis
21. Which statement about carbohydrates as energy sources is correct?
- A) Carbohydrates provide quick energy to cells ✓

- B) Carbohydrates cannot be oxidized
- C) Carbohydrates contain no calories
- D) Carbohydrates are used only for structure

22. Which statement correctly explains calorie value of carbohydrates?

- A) One gram of carbohydrate provides about 4 calories ✓
- B) One gram provides no energy
- C) Carbohydrates provide 9 calories per gram
- D) Calories are unrelated to carbohydrates

23. Which statement about dietary fiber is correct?

- A) Fiber helps maintain digestive health ✓
- B) Fiber is fully digested into glucose
- C) Fiber prevents intestinal movement
- D) Fiber contains proteins only

24. Which statement correctly describes soluble fiber?

- A) Soluble fiber dissolves in water ✓
- B) Soluble fiber cannot affect digestion
- C) Soluble fiber is a protein
- D) Soluble fiber prevents water absorption

25. Which statement about insoluble fiber is correct?

- A) Insoluble fiber adds bulk to stool ✓
- B) Insoluble fiber dissolves completely in water
- C) Insoluble fiber is a lipid
- D) Insoluble fiber prevents bowel movement

26. Which statement correctly explains dehydration synthesis in carbohydrates?

- A) Monosaccharides join by removal of water ✓
- B) Water is added to form bonds
- C) Dehydration destroys sugar molecules
- D) Dehydration synthesis occurs only in proteins

27. Which statement about hydrolysis is correct?

- A) Hydrolysis breaks carbohydrate bonds using water ✓
- B) Hydrolysis forms polysaccharides only
- C) Hydrolysis prevents digestion
- D) Hydrolysis contains no enzymes

28. Which statement correctly describes reducing sugars?

- A) Reducing sugars can donate electrons in reactions ✓
- B) Reducing sugars contain no oxygen
- C) Reducing sugars are proteins
- D) Reducing sugars cannot dissolve in water

29. Which statement about glucose storage is correct?

- A) Excess glucose may be stored as glycogen ✓
- B) Glucose cannot be stored

- C) Glucose is stored as proteins only
D) Glucose prevents metabolism
30. Which statement correctly explains blood glucose regulation?
A) Hormones help maintain stable blood glucose levels ✓
B) Blood glucose never changes
C) Glucose regulation prevents metabolism
D) Hormones destroy glucose permanently
31. Which statement about insulin is correct?
A) Insulin lowers blood glucose levels ✓
B) Insulin prevents glucose uptake
C) Insulin is a carbohydrate
D) Insulin stores oxygen
32. Which statement correctly describes glucagon?
A) Glucagon raises blood glucose levels ✓
B) Glucagon prevents glycogen breakdown
C) Glucagon is a lipid
D) Glucagon stops metabolism
33. Which statement about carbohydrate deficiency is correct?
A) Low carbohydrate intake may reduce energy availability ✓
B) Carbohydrate deficiency increases glucose storage
C) Carbohydrates are unnecessary in diet
D) Carbohydrate deficiency improves fatigue resistance
34. Which statement correctly explains excess carbohydrate intake?
A) Excess carbohydrates may be stored as fat ✓
B) Excess carbohydrates cannot affect the body
C) Carbohydrates prevent weight gain
D) Excess carbohydrates stop ATP production
35. Which statement about lactose intolerance is correct?
A) Lactose intolerance results from low lactase enzyme activity ✓
B) Lactose intolerance improves digestion
C) Lactose intolerance prevents all sugar absorption
D) Lactose intolerance is caused by protein excess
36. Which statement correctly describes lactase?
A) Lactase digests lactose into simple sugars ✓
B) Lactase digests proteins only
C) Lactase prevents digestion
D) Lactase is a lipid molecule
37. Which statement about carbohydrate-rich foods is correct?
A) Bread, rice, and potatoes are rich in carbohydrates ✓
B) Carbohydrates are found only in meat
C) Carbohydrate foods contain no energy
D) Carbohydrate foods prevent respiration

38. Which statement correctly explains complex carbohydrates?

- A) Complex carbohydrates contain long sugar chains ✓
- B) Complex carbohydrates are always monosaccharides
- C) Complex carbohydrates contain amino acids
- D) Complex carbohydrates cannot provide energy

39. Which statement about simple carbohydrates is correct?

- A) Simple carbohydrates are rapidly digested sugars ✓
- B) Simple carbohydrates cannot raise blood glucose
- C) Simple carbohydrates are proteins
- D) Simple carbohydrates contain no carbon

40. Which statement correctly describes glycosidic bonds?

- A) Glycosidic bonds join sugar molecules together ✓
- B) Glycosidic bonds connect amino acids
- C) Glycosidic bonds prevent polysaccharide formation
- D) Glycosidic bonds contain no oxygen

41. Which statement about cellulose digestion in humans is correct?

- A) Humans cannot digest cellulose efficiently ✓
- B) Humans completely digest cellulose into glucose
- C) Cellulose digestion produces proteins
- D) Humans store cellulose as glycogen

42. Which statement correctly explains why cellulose is important in plants?

- A) Cellulose provides rigidity and support ✓
- B) Cellulose stores genetic information
- C) Cellulose transports oxygen
- D) Cellulose prevents growth

43. Which statement about glycogen storage is correct?

- A) Glycogen is mainly stored in liver and muscles ✓
- B) Glycogen is stored only in bones
- C) Glycogen cannot release glucose
- D) Glycogen prevents ATP production

44. Which statement correctly describes carbohydrate oxidation?

- A) Oxidation of glucose releases energy ✓
- B) Oxidation prevents ATP formation
- C) Oxidation occurs only in plants
- D) Oxidation destroys all enzymes

45. Which statement about fermentation is correct?

- A) Fermentation allows energy production without oxygen ✓
- B) Fermentation requires high oxygen levels
- C) Fermentation produces no ATP
- D) Fermentation occurs only in plants

46. Which statement correctly explains the role of carbohydrates in exercise?

- A) Carbohydrates provide energy during physical activity ✓

- B) Carbohydrates weaken muscles permanently
 - C) Carbohydrates stop respiration
 - D) Carbohydrates prevent ATP production
47. Which statement about carbohydrate loading is correct?
- A) Athletes may increase carbohydrate intake to store glycogen ✓
 - B) Carbohydrate loading reduces energy reserves
 - C) Carbohydrate loading prevents endurance
 - D) Carbohydrate loading removes glucose from muscles
48. Which statement correctly describes pentose sugars?
- A) Pentose sugars contain five carbon atoms ✓
 - B) Pentose sugars contain six carbon atoms
 - C) Pentose sugars are proteins
 - D) Pentose sugars cannot form nucleic acids
49. Which statement about ribose is correct?
- A) Ribose is a sugar found in RNA ✓
 - B) Ribose is a lipid molecule
 - C) Ribose prevents nucleotide formation
 - D) Ribose stores oxygen
50. Which statement correctly describes deoxyribose?
- A) Deoxyribose is a sugar present in DNA ✓
 - B) Deoxyribose is a protein
 - C) Deoxyribose cannot form nucleotides
 - D) Deoxyribose stores ATP
51. Which statement about hexose sugars is correct?
- A) Hexose sugars contain six carbon atoms ✓
 - B) Hexose sugars contain three carbon atoms
 - C) Hexose sugars are proteins only
 - D) Hexose sugars contain no oxygen
52. Which statement correctly explains energy release from carbohydrates?
- A) Energy is released when glucose bonds are broken ✓
 - B) Carbohydrates absorb all ATP
 - C) Carbohydrates cannot undergo respiration
 - D) Energy release destroys enzymes permanently
53. Which statement about starch digestion is correct?
- A) Starch is broken down into glucose units ✓
 - B) Starch digestion produces amino acids
 - C) Starch cannot be hydrolyzed
 - D) Starch digestion stops metabolism
54. Which statement correctly describes oligosaccharides?
- A) Oligosaccharides contain a few monosaccharide units ✓
 - B) Oligosaccharides contain hundreds of sugars only

- C) Oligosaccharides are proteins
- D) Oligosaccharides prevent digestion

55. Which statement about carbohydrate functions is correct?

- A) Carbohydrates provide energy and structural support ✓
- B) Carbohydrates store genetic information only
- C) Carbohydrates prevent metabolism
- D) Carbohydrates are unnecessary in cells

56. Which statement correctly explains why glucose is important for the brain?

- A) The brain relies heavily on glucose for energy ✓
- B) The brain cannot use glucose
- C) Glucose damages nerve cells permanently
- D) The brain stores only proteins for energy

57. Which statement about energy reserves is correct?

- A) Glycogen acts as a short-term energy reserve ✓
- B) Glycogen stores oxygen only
- C) Glycogen cannot release glucose
- D) Glycogen prevents ATP production

58. Which statement correctly describes carbohydrate absorption?

- A) Simple sugars are absorbed through the intestine ✓
- B) Carbohydrates are absorbed without digestion
- C) Polysaccharides enter blood directly
- D) Absorption prevents metabolism

59. Which statement about glucose transport is correct?

- A) Blood transports glucose to body cells ✓
- B) Glucose cannot circulate in blood
- C) Glucose transport occurs only in plants
- D) Blood destroys glucose permanently

60. Which statement correctly explains carbohydrate importance in plants?

- A) Plants produce carbohydrates through photosynthesis ✓
- B) Plants cannot synthesize glucose
- C) Plants use proteins as primary photosynthetic products
- D) Plants prevent carbohydrate formation

61. Which statement about photosynthesis is correct?

- A) Photosynthesis converts light energy into glucose ✓
- B) Photosynthesis destroys carbohydrates
- C) Photosynthesis occurs only in animals
- D) Photosynthesis prevents ATP formation

62. Which statement correctly describes carbon fixation?

- A) Carbon dioxide is incorporated into carbohydrates ✓
- B) Carbon fixation removes glucose from plants
- C) Carbon fixation prevents photosynthesis
- D) Carbon fixation occurs only in animals

63. Which statement about glycemic index is correct?
A) Glycemic index measures how quickly foods raise blood glucose ✓
B) Glycemic index measures protein digestion only
C) Glycemic index has no nutritional value
D) Glycemic index prevents glucose absorption
64. Which statement correctly explains high glycemic foods?
A) They raise blood glucose rapidly ✓
B) They cannot provide energy
C) They contain no carbohydrates
D) They reduce glucose absorption permanently
65. Which statement about low glycemic foods is correct?
A) They release glucose more gradually ✓
B) They contain no energy
C) They stop carbohydrate digestion
D) They cannot affect blood sugar
66. Which statement correctly describes sucrose digestion?
A) Sucrose is broken into glucose and fructose ✓
B) Sucrose digestion produces amino acids
C) Sucrose remains undigested always
D) Sucrose digestion prevents ATP production
67. Which statement about carbohydrate storage in plants is correct?
A) Plants store excess glucose as starch ✓
B) Plants store glucose only as proteins
C) Plants cannot store carbohydrates
D) Starch prevents energy storage
68. Which statement correctly explains why athletes consume carbohydrates?
A) Carbohydrates replenish glycogen and provide energy ✓
B) Carbohydrates weaken muscles
C) Carbohydrates stop respiration
D) Carbohydrates destroy ATP
69. Which statement about cellulose fibers is correct?
A) Cellulose fibers strengthen plant tissues ✓
B) Cellulose fibers transport oxygen
C) Cellulose fibers are proteins
D) Cellulose fibers prevent support
70. Which statement correctly describes carbohydrate chemical composition?
A) Carbohydrates generally contain hydrogen and oxygen in a 2:1 ratio ✓
B) Carbohydrates contain sulfur only
C) Carbohydrates contain no carbon
D) Carbohydrates lack oxygen
71. Which statement about glucose solubility is correct?
A) Glucose dissolves easily in water ✓

- B) Glucose is completely insoluble
- C) Glucose is a lipid
- D) Glucose prevents hydration

72. Which statement correctly explains glycogen breakdown?

- A) Glycogen can be converted back into glucose ✓
- B) Glycogen cannot release energy
- C) Glycogen breakdown forms proteins
- D) Glycogen breakdown stops metabolism

73. Which statement about carbohydrate catabolism is correct?

- A) Catabolism breaks carbohydrates into simpler molecules ✓
- B) Catabolism builds glycogen only
- C) Catabolism prevents ATP release
- D) Catabolism occurs only in plants

74. Which statement correctly describes anabolic carbohydrate reactions?

- A) Anabolic reactions build complex carbohydrates ✓
- B) Anabolic reactions destroy glucose only
- C) Anabolic reactions stop metabolism
- D) Anabolic reactions release no energy

75. Which statement about metabolic pathways is correct?

- A) Carbohydrates participate in many metabolic pathways ✓
- B) Carbohydrates are metabolically inactive
- C) Carbohydrates cannot form ATP
- D) Carbohydrates prevent enzyme activity

76. Which statement correctly explains the sweetness of sugars?

- A) Many monosaccharides and disaccharides taste sweet ✓
- B) All carbohydrates are tasteless
- C) Polysaccharides are always sweeter than monosaccharides
- D) Sweetness prevents digestion

77. Which statement about glucose homeostasis is correct?

- A) The body maintains stable glucose concentration ✓
- B) Glucose levels never change
- C) Homeostasis prevents glucose uptake
- D) Glucose balance is unimportant

78. Which statement correctly describes energy storage efficiency of carbohydrates?

- A) Carbohydrates provide readily available energy ✓
- B) Carbohydrates cannot be stored
- C) Carbohydrates are slower than all nutrients for energy
- D) Carbohydrates prevent ATP synthesis

79. Which statement about starch granules is correct?

- A) Plants store starch in granule form ✓
- B) Starch granules contain proteins only

- C) Starch granules prevent energy storage
- D) Starch granules exist only in animals

80. Which statement correctly explains why glucose is called blood sugar?

- A) Glucose circulates in the bloodstream ✓
- B) Glucose exists only in bones
- C) Glucose prevents circulation
- D) Glucose contains amino acids

81. Which statement about energy transformation is correct?

- A) Cells transform carbohydrate energy into ATP ✓
- B) ATP cannot store energy
- C) Carbohydrates destroy mitochondria
- D) Energy transformation stops metabolism

82. Which statement correctly describes starch structure?

- A) Starch consists of many glucose molecules ✓
- B) Starch contains amino acids
- C) Starch is a monosaccharide
- D) Starch cannot form polymers

83. Which statement about glycogen structure is correct?

- A) Glycogen is a highly branched glucose polymer ✓
- B) Glycogen contains proteins only
- C) Glycogen is a disaccharide
- D) Glycogen cannot release glucose

84. Which statement correctly explains cellulose structure?

- A) Cellulose consists of long chains of glucose ✓
- B) Cellulose contains amino acids
- C) Cellulose is a lipid
- D) Cellulose cannot provide support

85. Which statement about carbohydrate importance in cellular respiration is correct?

- A) Glucose is a primary fuel in respiration ✓
- B) Carbohydrates prevent ATP production
- C) Respiration cannot use carbohydrates
- D) Carbohydrates stop electron transport

86. Which statement correctly describes dietary carbohydrates?

- A) Dietary carbohydrates are important for energy supply ✓
- B) Dietary carbohydrates have no biological role
- C) Dietary carbohydrates cannot be digested
- D) Dietary carbohydrates stop metabolism

87. Which statement about sugar metabolism disorders is correct?

- A) Abnormal glucose regulation can lead to disease ✓
- B) Sugar metabolism never affects health
- C) Glucose disorders improve ATP production
- D) Sugar metabolism is unrelated to hormones

88. Which statement correctly explains diabetes mellitus?
- A) Diabetes involves impaired glucose regulation ✓
 - B) Diabetes prevents carbohydrate digestion only
 - C) Diabetes occurs without insulin involvement
 - D) Diabetes increases unlimited glucose uptake
89. Which statement about carbohydrate reserves during fasting is correct?
- A) Glycogen reserves help maintain blood glucose ✓
 - B) Glycogen cannot be used during fasting
 - C) Fasting increases glycogen formation only
 - D) Glycogen prevents metabolism
90. Which statement correctly describes carbohydrate oxidation products?
- A) Oxidation produces carbon dioxide and water ✓
 - B) Oxidation produces proteins only
 - C) Oxidation prevents ATP formation
 - D) Oxidation contains no oxygen
91. Which statement about energy needs is correct?
- A) Carbohydrates help meet daily energy requirements ✓
 - B) Carbohydrates cannot supply calories
 - C) Energy needs are unrelated to glucose
 - D) Carbohydrates stop cellular work
92. Which statement correctly explains the role of enzymes in carbohydrate digestion?
- A) Enzymes speed breakdown of carbohydrates ✓
 - B) Enzymes prevent digestion
 - C) Enzymes are carbohydrates only
 - D) Enzymes stop glucose absorption
93. Which statement about carbohydrate biosynthesis is correct?
- A) Organisms can synthesize carbohydrates from smaller molecules ✓
 - B) Carbohydrates cannot be synthesized
 - C) Biosynthesis destroys glucose permanently
 - D) Biosynthesis occurs only in animals
94. Which statement correctly describes glucose as a monosaccharide?
- A) Glucose is a single sugar unit ✓
 - B) Glucose is a polysaccharide
 - C) Glucose contains proteins
 - D) Glucose cannot dissolve in water
95. Which statement about carbohydrate diversity is correct?
- A) Carbohydrates vary in size and function ✓
 - B) All carbohydrates are identical
 - C) Carbohydrates contain no structural differences
 - D) Carbohydrates cannot form polymers
96. Which statement correctly explains energy release during exercise?
- A) Muscles use glucose and glycogen for ATP production ✓

- B) Exercise prevents carbohydrate metabolism
- C) Muscles cannot use glucose
- D) ATP is unrelated to carbohydrates

97. Which statement about starch hydrolysis is correct?

- A) Hydrolysis converts starch into smaller sugars ✓
- B) Hydrolysis forms proteins from starch
- C) Hydrolysis prevents glucose release
- D) Hydrolysis occurs without water

98. Which statement correctly describes carbohydrates in living organisms?

- A) Carbohydrates are essential biomolecules in cells ✓
- B) Carbohydrates exist only outside organisms
- C) Carbohydrates cannot support life
- D) Carbohydrates contain no carbon

99. Which statement about carbohydrate advantages is correct?

- A) Carbohydrates provide rapid and accessible energy ✓
- B) Carbohydrates have no biological value
- C) Carbohydrates prevent metabolism
- D) Carbohydrates destroy ATP

100. Which statement correctly explains the biological importance of carbohydrates?

- A) Carbohydrates support energy production, storage, and structure ✓
- B) Carbohydrates contain no functional importance
- C) Carbohydrates are unnecessary for cells
- D) Carbohydrates stop biochemical reactions

1. Which statement best defines lipids?

- A) Lipids are water-soluble proteins
- B) Lipids are organic compounds insoluble in water ✓
- C) Lipids contain only nitrogen
- D) Lipids are inorganic salts

2. Which statement correctly describes the main function of lipids?

- A) Lipids primarily store long-term energy ✓
- B) Lipids prevent energy storage
- C) Lipids store genetic information
- D) Lipids contain no calories

3. Which statement about fatty acids is correct?

- A) Fatty acids are components of many lipids ✓
- B) Fatty acids contain amino acids only
- C) Fatty acids cannot form triglycerides
- D) Fatty acids are proteins

4. Which statement correctly describes saturated fatty acids?

- A) Saturated fatty acids contain no double bonds ✓
- B) Saturated fatty acids contain many double bonds

- C) Saturated fatty acids are always liquid
D) Saturated fatty acids contain no carbon
5. Which statement about unsaturated fatty acids is correct?
A) Unsaturated fatty acids contain one or more double bonds ✓
B) Unsaturated fatty acids are fully saturated with hydrogen
C) Unsaturated fatty acids are proteins
D) Unsaturated fatty acids cannot store energy
6. Which statement correctly describes monounsaturated fatty acids?
A) They contain one double bond ✓
B) They contain no double bonds
C) They contain many peptide bonds
D) They are carbohydrates
7. Which statement about polyunsaturated fatty acids is correct?
A) They contain multiple double bonds ✓
B) They contain only one carbon atom
C) They are saturated lipids
D) They cannot participate in metabolism
8. Which statement correctly explains triglycerides?
A) Triglycerides consist of glycerol and three fatty acids ✓
B) Triglycerides are proteins only
C) Triglycerides contain no glycerol
D) Triglycerides prevent energy storage
9. Which statement about glycerol is correct?
A) Glycerol forms the backbone of triglycerides ✓
B) Glycerol is an amino acid
C) Glycerol prevents lipid formation
D) Glycerol is a polysaccharide
10. Which statement correctly describes phospholipids?
A) Phospholipids are major components of cell membranes ✓
B) Phospholipids contain no fatty acids
C) Phospholipids are proteins
D) Phospholipids prevent membrane formation
11. Which statement about phospholipid structure is correct?
A) Phospholipids have hydrophilic heads and hydrophobic tails ✓
B) Phospholipids are completely water-soluble
C) Phospholipids contain amino acids only
D) Phospholipids lack phosphate groups
12. Which statement correctly explains steroids?
A) Steroids are lipids with four fused carbon rings ✓
B) Steroids are polysaccharides
C) Steroids contain no carbon
D) Steroids prevent hormone formation

13. Which statement about cholesterol is correct?
A) Cholesterol is an important steroid in animal cells ✓
B) Cholesterol is a carbohydrate
C) Cholesterol prevents membrane stability
D) Cholesterol contains amino acids
14. Which statement correctly describes waxes?
A) Waxes provide waterproof protection ✓
B) Waxes are proteins only
C) Waxes dissolve easily in water
D) Waxes contain glucose polymers
15. Which statement about lipid solubility is correct?
A) Lipids dissolve in nonpolar solvents ✓
B) Lipids dissolve completely in water
C) Lipids contain no hydrophobic regions
D) Lipids prevent solvent interactions
16. Which statement correctly explains essential fatty acids?
A) Essential fatty acids must be obtained from diet ✓
B) Essential fatty acids are produced in unlimited amounts by the body
C) Essential fatty acids contain no carbon
D) Essential fatty acids are proteins
17. Which statement about omega-3 fatty acids is correct?
A) Omega-3 fatty acids are beneficial unsaturated fats ✓
B) Omega-3 fatty acids are saturated fats only
C) Omega-3 fatty acids prevent metabolism
D) Omega-3 fatty acids are carbohydrates
18. Which statement correctly describes omega-6 fatty acids?
A) Omega-6 fatty acids are essential polyunsaturated fats ✓
B) Omega-6 fatty acids contain no double bonds
C) Omega-6 fatty acids are proteins
D) Omega-6 fatty acids prevent growth
19. Which statement about trans fats is correct?
A) Trans fats are associated with health risks ✓
B) Trans fats improve all metabolic functions
C) Trans fats are essential nutrients
D) Trans fats contain amino acids only
20. Which statement correctly explains lipid digestion?
A) Lipids are broken down into fatty acids and glycerol ✓
B) Lipids cannot be digested
C) Lipid digestion forms proteins
D) Lipid digestion prevents energy release
21. Which statement about bile salts is correct?
A) Bile salts help emulsify fats ✓

- B) Bile salts digest proteins only
- C) Bile salts prevent lipid absorption
- D) Bile salts are carbohydrates

22. Which statement correctly describes lipase?

- A) Lipase is an enzyme that digests lipids ✓
- B) Lipase digests proteins only
- C) Lipase prevents fatty acid formation
- D) Lipase is a phospholipid

23. Which statement about adipose tissue is correct?

- A) Adipose tissue stores fats in the body ✓
- B) Adipose tissue stores proteins only
- C) Adipose tissue prevents insulation
- D) Adipose tissue contains no lipids

24. Which statement correctly explains the insulation function of fats?

- A) Fats help reduce heat loss ✓
- B) Fats prevent temperature regulation
- C) Fats contain no energy
- D) Fats destroy body tissues

25. Which statement about lipid energy content is correct?

- A) Lipids provide about 9 calories per gram ✓
- B) Lipids provide no energy
- C) Lipids provide 4 calories per gram only
- D) Lipids cannot be metabolized

26. Which statement correctly describes phospholipid bilayers?

- A) Phospholipids form bilayers in cell membranes ✓
- B) Phospholipids form DNA strands
- C) Phospholipids prevent membrane formation
- D) Phospholipids are water-soluble proteins

27. Which statement about hydrophobic lipid tails is correct?

- A) Hydrophobic tails repel water ✓
- B) Hydrophobic tails attract water strongly
- C) Hydrophobic tails are carbohydrates
- D) Hydrophobic tails contain phosphate groups only

28. Which statement correctly explains lipid metabolism?

- A) Lipids can be broken down to release energy ✓
- B) Lipids cannot participate in respiration
- C) Lipids stop ATP production
- D) Lipids are metabolically inactive

29. Which statement about lipoproteins is correct?

- A) Lipoproteins transport lipids in blood ✓
- B) Lipoproteins contain no proteins

- C) Lipoproteins prevent circulation
- D) Lipoproteins are polysaccharides

30. Which statement correctly describes HDL?

- A) HDL helps transport excess cholesterol away from tissues ✓
- B) HDL stores glucose only
- C) HDL is a triglyceride
- D) HDL prevents lipid transport

31. Which statement about LDL is correct?

- A) High LDL levels may increase heart disease risk ✓
- B) LDL prevents cholesterol transport
- C) LDL is a carbohydrate
- D) LDL contains no lipids

32. Which statement correctly explains lipid storage?

- A) Lipids are stored efficiently due to high energy content ✓
- B) Lipids cannot be stored in cells
- C) Lipids store less energy than carbohydrates
- D) Lipids prevent ATP synthesis

33. Which statement about saturated fats is correct?

- A) Saturated fats are usually solid at room temperature ✓
- B) Saturated fats are always liquid
- C) Saturated fats contain many double bonds
- D) Saturated fats are proteins

34. Which statement correctly describes unsaturated fats?

- A) Unsaturated fats are usually liquid at room temperature ✓
- B) Unsaturated fats contain no double bonds
- C) Unsaturated fats are always solid
- D) Unsaturated fats prevent metabolism

35. Which statement about phospholipid heads is correct?

- A) Phospholipid heads are hydrophilic ✓
- B) Phospholipid heads repel water completely
- C) Phospholipid heads contain fatty acids only
- D) Phospholipid heads are insoluble in water

36. Which statement correctly explains the role of cholesterol in membranes?

- A) Cholesterol helps maintain membrane fluidity ✓
- B) Cholesterol destroys membranes
- C) Cholesterol prevents cell communication
- D) Cholesterol is absent from animal membranes

37. Which statement about steroid hormones is correct?

- A) Some hormones are derived from lipids ✓
- B) Steroid hormones are proteins only
- C) Steroid hormones contain no carbon
- D) Steroid hormones prevent metabolism

38. Which statement correctly describes testosterone?

- A) Testosterone is a steroid hormone ✓
- B) Testosterone is a carbohydrate
- C) Testosterone prevents growth
- D) Testosterone contains amino acids only

39. Which statement about estrogen is correct?

- A) Estrogen is a lipid-derived hormone ✓
- B) Estrogen is a triglyceride
- C) Estrogen prevents reproduction
- D) Estrogen is a polysaccharide

40. Which statement correctly explains lipid advantages?

- A) Lipids provide energy, insulation, and protection ✓
- B) Lipids have no biological role
- C) Lipids prevent metabolism
- D) Lipids destroy cell membranes

41. Which statement about wax coatings in plants is correct?

- A) Waxes help reduce water loss ✓
- B) Waxes increase dehydration
- C) Waxes prevent structural support
- D) Waxes are proteins only

42. Which statement correctly describes subcutaneous fat?

- A) Subcutaneous fat lies beneath the skin ✓
- B) Subcutaneous fat stores oxygen only
- C) Subcutaneous fat contains no lipids
- D) Subcutaneous fat prevents insulation

43. Which statement about lipid hydrolysis is correct?

- A) Hydrolysis breaks lipids into smaller molecules ✓
- B) Hydrolysis forms triglycerides only
- C) Hydrolysis prevents digestion
- D) Hydrolysis contains no water

44. Which statement correctly explains dehydration synthesis in lipids?

- A) Water is removed to form lipid bonds ✓
- B) Water is added to create triglycerides
- C) Dehydration destroys fatty acids
- D) Dehydration synthesis occurs only in proteins

45. Which statement about energy reserves is correct?

- A) Lipids act as long-term energy reserves ✓
- B) Lipids cannot store energy
- C) Lipids are used only for structure
- D) Lipids prevent ATP production

46. Which statement correctly describes phospholipid function?

- A) Phospholipids help form selective barriers in membranes ✓

- B) Phospholipids digest proteins
- C) Phospholipids store genetic information
- D) Phospholipids prevent transport

47. Which statement about ketone bodies is correct?

- A) Ketone bodies may form during fat metabolism ✓
- B) Ketone bodies are proteins only
- C) Ketone bodies prevent energy release
- D) Ketone bodies are structural carbohydrates

48. Which statement correctly explains lipid transport in the body?

- A) Lipids are transported with the help of lipoproteins ✓
- B) Lipids dissolve freely in blood
- C) Lipids cannot circulate in the body
- D) Lipids prevent circulation

49. Which statement about fatty acid chains is correct?

- A) Fatty acid chains vary in length and saturation ✓
- B) All fatty acids are identical
- C) Fatty acids contain no carbon atoms
- D) Fatty acids cannot form triglycerides

50. Which statement correctly explains the biological importance of lipids?

- A) Lipids are essential for energy storage, membranes, and signaling ✓
- B) Lipids have no cellular functions
- C) Lipids prevent biochemical reactions
- D) Lipids are unnecessary biomolecules

1. Which statement correctly describes prokaryotic cells?

- A) Prokaryotic cells lack a membrane-bound nucleus ✓
- B) Prokaryotic cells contain membrane-bound organelles
- C) Prokaryotic cells are always multicellular
- D) Prokaryotic cells contain no ribosomes

2. Which statement about eukaryotic cells is correct?

- A) Eukaryotic cells contain a true nucleus ✓
- B) Eukaryotic cells lack DNA
- C) Eukaryotic cells are always unicellular
- D) Eukaryotic cells contain no organelles

3. Which statement correctly explains the main difference between prokaryotic and eukaryotic cells?

- A) Eukaryotes possess membrane-bound organelles ✓
- B) Prokaryotes contain larger nuclei
- C) Eukaryotes lack ribosomes
- D) Prokaryotes cannot reproduce

4. Which statement about bacterial cells is correct?

- A) Bacteria are prokaryotic organisms ✓
- B) Bacteria contain mitochondria

- C) Bacteria have membrane-bound nuclei
D) Bacteria are eukaryotes
5. Which statement correctly describes the nucleus in eukaryotic cells?
A) The nucleus stores genetic material ✓
B) The nucleus produces ATP only
C) The nucleus lacks DNA
D) The nucleus is absent in all cells
6. Which statement about ribosomes is correct?
A) Ribosomes synthesize proteins ✓
B) Ribosomes store lipids
C) Ribosomes contain no RNA
D) Ribosomes prevent translation
7. Which statement correctly describes the nucleoid region?
A) It contains DNA in prokaryotic cells ✓
B) It is a membrane-bound organelle
C) It contains proteins only
D) It prevents genetic activity
8. Which statement about organelles is correct?
A) Membrane-bound organelles are characteristic of eukaryotes ✓
B) Organelles are absent in eukaryotes
C) Organelles contain no enzymes
D) Organelles exist only in bacteria
9. Which statement correctly explains DNA?
A) DNA stores hereditary information ✓
B) DNA is a carbohydrate
C) DNA prevents protein synthesis
D) DNA contains amino acids only
10. Which statement about RNA is correct?
A) RNA plays a role in protein synthesis ✓
B) RNA contains thymine instead of uracil
C) RNA is double-stranded always
D) RNA stores fats in cells
11. Which statement correctly describes nucleic acids?
A) Nucleic acids include DNA and RNA ✓
B) Nucleic acids are proteins only
C) Nucleic acids contain no nucleotides
D) Nucleic acids prevent inheritance
12. Which statement about nucleotides is correct?
A) Nucleotides are building blocks of nucleic acids ✓
B) Nucleotides are amino acids
C) Nucleotides contain no phosphate groups
D) Nucleotides cannot form DNA

13. Which statement correctly describes the structure of a nucleotide?

- A) A nucleotide contains sugar, phosphate, and nitrogenous base ✓
- B) A nucleotide contains amino acids only
- C) A nucleotide lacks carbon atoms
- D) A nucleotide is a lipid

14. Which statement about adenine is correct?

- A) Adenine is a nitrogenous base found in DNA and RNA ✓
- B) Adenine is a phospholipid
- C) Adenine prevents replication
- D) Adenine is a protein

15. Which statement correctly explains base pairing in DNA?

- A) Adenine pairs with thymine ✓
- B) Adenine pairs with cytosine
- C) Guanine pairs with thymine
- D) Cytosine pairs with adenine only

16. Which statement about guanine and cytosine is correct?

- A) Guanine pairs with cytosine ✓
- B) Guanine pairs with thymine
- C) Cytosine pairs with adenine
- D) Guanine prevents replication

17. Which statement correctly describes RNA bases?

- A) RNA contains uracil instead of thymine ✓
- B) RNA contains no nitrogenous bases
- C) RNA contains thymine only
- D) RNA prevents protein synthesis

18. Which statement about DNA structure is correct?

- A) DNA is a double helix ✓
- B) DNA is always single-stranded
- C) DNA lacks phosphate groups
- D) DNA is a polysaccharide

19. Which statement correctly explains DNA replication?

- A) DNA replication produces identical DNA molecules ✓
- B) DNA replication destroys nucleotides
- C) DNA replication occurs only in bacteria
- D) DNA replication prevents inheritance

20. Which statement about messenger RNA is correct?

- A) mRNA carries genetic instructions for protein synthesis ✓
- B) mRNA stores lipids
- C) mRNA prevents translation
- D) mRNA contains no nucleotides

21. Which statement correctly describes transfer RNA?

- A) tRNA carries amino acids during protein synthesis ✓

- B) tRNA stores ATP only
- C) tRNA prevents ribosome function
- D) tRNA contains no anticodons

22. Which statement about ribosomal RNA is correct?

- A) rRNA forms part of ribosome structure ✓
- B) rRNA stores genetic information only
- C) rRNA prevents translation
- D) rRNA is a lipid molecule

23. Which statement correctly explains transcription?

- A) Transcription produces RNA from DNA ✓
- B) Transcription forms proteins directly
- C) Transcription prevents gene expression
- D) Transcription occurs outside cells only

24. Which statement about translation is correct?

- A) Translation synthesizes proteins from mRNA ✓
- B) Translation produces DNA only
- C) Translation destroys amino acids
- D) Translation prevents protein formation

25. Which statement correctly describes chromosomes?

- A) Chromosomes contain DNA and proteins ✓
- B) Chromosomes contain carbohydrates only
- C) Chromosomes prevent inheritance
- D) Chromosomes lack genes

26. Which statement about genes is correct?

- A) Genes are segments of DNA ✓
- B) Genes are phospholipids
- C) Genes prevent protein synthesis
- D) Genes contain no hereditary information

27. Which statement correctly explains mutations?

- A) Mutations are changes in DNA sequence ✓
- B) Mutations always improve organisms
- C) Mutations prevent inheritance
- D) Mutations occur only in RNA

28. Which statement about plasma membrane is correct?

- A) The plasma membrane surrounds the cell ✓
- B) The plasma membrane stores DNA
- C) The plasma membrane contains no proteins
- D) The plasma membrane is absent in cells

29. Which statement correctly describes the cell membrane composition?

- A) The membrane mainly consists of phospholipids and proteins ✓
- B) The membrane contains only carbohydrates

- C) The membrane lacks lipids
- D) The membrane contains DNA only

30. Which statement about the fluid mosaic model is correct?

- A) The membrane contains moving lipids and proteins ✓
- B) The membrane is completely rigid
- C) The membrane lacks organization
- D) The membrane contains no phospholipids

31. Which statement correctly explains phospholipid bilayers?

- A) Phospholipids arrange into a bilayer in membranes ✓
- B) Phospholipids form protein chains
- C) Phospholipids prevent membrane formation
- D) Phospholipids are carbohydrates

32. Which statement about hydrophilic heads is correct?

- A) Hydrophilic heads interact with water ✓
- B) Hydrophilic heads repel water
- C) Hydrophilic heads contain no phosphate groups
- D) Hydrophilic heads are proteins only

33. Which statement correctly describes hydrophobic tails?

- A) Hydrophobic tails avoid water ✓
- B) Hydrophobic tails dissolve easily in water
- C) Hydrophobic tails are carbohydrates
- D) Hydrophobic tails contain amino acids only

34. Which statement about membrane proteins is correct?

- A) Membrane proteins assist transport and signaling ✓
- B) Membrane proteins contain no amino acids
- C) Membrane proteins prevent communication
- D) Membrane proteins are absent in plasma membranes

35. Which statement correctly explains selective permeability?

- A) The membrane controls movement of substances ✓
- B) The membrane allows all substances equally
- C) The membrane blocks all molecules permanently
- D) The membrane prevents transport completely

36. Which statement about diffusion is correct?

- A) Diffusion moves substances from high to low concentration ✓
- B) Diffusion requires ATP always
- C) Diffusion prevents transport
- D) Diffusion occurs only in solids

37. Which statement correctly describes osmosis?

- A) Osmosis is diffusion of water across a membrane ✓
- B) Osmosis transports proteins only
- C) Osmosis prevents water movement
- D) Osmosis occurs without membranes

38. Which statement about active transport is correct?

- A) Active transport requires energy ✓
- B) Active transport occurs without ATP
- C) Active transport moves substances only down gradients
- D) Active transport prevents membrane function

39. Which statement correctly explains channel proteins?

- A) Channel proteins allow specific substances to pass ✓
- B) Channel proteins digest carbohydrates
- C) Channel proteins store DNA
- D) Channel proteins prevent transport

40. Which statement about carrier proteins is correct?

- A) Carrier proteins transport molecules across membranes ✓
- B) Carrier proteins contain no proteins
- C) Carrier proteins prevent diffusion
- D) Carrier proteins are polysaccharides

41. Which statement correctly describes receptor proteins?

- A) Receptor proteins receive chemical signals ✓
- B) Receptor proteins digest lipids
- C) Receptor proteins store ATP
- D) Receptor proteins prevent communication

42. Which statement about cholesterol in membranes is correct?

- A) Cholesterol helps maintain membrane fluidity ✓
- B) Cholesterol destroys phospholipids
- C) Cholesterol prevents transport
- D) Cholesterol is absent from animal membranes

43. Which statement correctly explains glycoproteins in membranes?

- A) Glycoproteins assist in cell recognition ✓
- B) Glycoproteins contain no carbohydrates
- C) Glycoproteins prevent signaling
- D) Glycoproteins store oxygen

44. Which statement about glycolipids is correct?

- A) Glycolipids contain carbohydrate chains attached to lipids ✓
- B) Glycolipids are proteins only
- C) Glycolipids prevent membrane stability
- D) Glycolipids contain no carbon

45. Which statement correctly describes endocytosis?

- A) Endocytosis brings materials into the cell ✓
- B) Endocytosis exports substances only
- C) Endocytosis prevents membrane movement
- D) Endocytosis occurs without energy

46. Which statement about exocytosis is correct?

- A) Exocytosis releases materials outside the cell ✓

- B) Exocytosis prevents secretion
- C) Exocytosis imports nutrients only
- D) Exocytosis occurs without vesicles

47. Which statement correctly explains plasma membrane importance?

- A) The plasma membrane maintains cellular homeostasis ✓
- B) The plasma membrane prevents all interactions
- C) The plasma membrane stores hereditary information
- D) The plasma membrane contains no lipids

48. Which statement about prokaryotic DNA is correct?

- A) Prokaryotic DNA is usually circular ✓
- B) Prokaryotic DNA is always linear
- C) Prokaryotic DNA is absent
- D) Prokaryotic DNA cannot replicate

49. Which statement correctly describes eukaryotic DNA?

- A) Eukaryotic DNA is organized into linear chromosomes ✓
- B) Eukaryotic DNA is always circular
- C) Eukaryotic DNA lacks genes
- D) Eukaryotic DNA prevents inheritance

50. Which statement correctly explains the biological importance of nucleic acids and membranes?

- A) Nucleic acids store information while membranes regulate transport ✓
- B) Nucleic acids prevent inheritance and membranes stop metabolism
- C) Membranes store genes and nucleic acids form barriers
- D) Nucleic acids and membranes have no cellular importance

1. Which statement correctly describes the fluid mosaic model?

- A) The membrane consists of moving lipids and proteins ✓
- B) The membrane is completely rigid
- C) The membrane contains only carbohydrates
- D) The membrane lacks organization

2. Which statement about phospholipids in the fluid mosaic model is correct?

- A) Phospholipids form a flexible bilayer ✓
- B) Phospholipids contain amino acids only
- C) Phospholipids prevent membrane formation
- D) Phospholipids are carbohydrates

3. Which statement correctly explains membrane fluidity?

- A) Lipids and proteins can move within the membrane ✓
- B) Membrane components are permanently fixed
- C) Membrane fluidity prevents transport
- D) Membrane fluidity destroys proteins

4. Which statement about membrane proteins is correct?

- A) Proteins are embedded within the phospholipid bilayer ✓
- B) Proteins exist only outside membranes

- C) Proteins prevent communication
D) Proteins contain no amino acids
5. Which statement correctly describes cholesterol in membranes?
A) Cholesterol helps maintain membrane flexibility ✓
B) Cholesterol destroys phospholipids
C) Cholesterol prevents transport
D) Cholesterol is absent from animal membranes
6. Which statement about glycoproteins is correct?
A) Glycoproteins help in cell recognition ✓
B) Glycoproteins contain no carbohydrates
C) Glycoproteins prevent signaling
D) Glycoproteins are lipids only
7. Which statement correctly explains glycolipids?
A) Glycolipids contain carbohydrate chains attached to lipids ✓
B) Glycolipids are proteins
C) Glycolipids prevent membrane stability
D) Glycolipids contain amino acids only
8. Which statement about the plasma membrane is correct?
A) The plasma membrane surrounds and protects the cell ✓
B) The plasma membrane stores DNA
C) The plasma membrane contains no lipids
D) The plasma membrane is absent in cells
9. Which statement correctly describes selective permeability?
A) The membrane controls entry and exit of substances ✓
B) The membrane allows all substances equally
C) The membrane blocks all molecules
D) The membrane prevents transport completely
10. Which statement about the cell wall is correct?
A) The cell wall provides support and protection ✓
B) The cell wall is found only in animal cells
C) The cell wall stores genetic information
D) The cell wall replaces the plasma membrane completely
11. Which statement correctly describes plant cell walls?
A) Plant cell walls mainly contain cellulose ✓
B) Plant cell walls contain proteins only
C) Plant cell walls are made of glycogen
D) Plant cell walls prevent rigidity
12. Which statement about bacterial cell walls is correct?
A) Bacterial cell walls contain peptidoglycan ✓
B) Bacterial cell walls contain cellulose only
C) Bacterial cell walls are absent
D) Bacterial cell walls contain DNA only

13. Which statement correctly explains fungal cell walls?
- A) Fungal cell walls contain chitin ✓
 - B) Fungal cell walls contain glycogen only
 - C) Fungal cell walls are made of proteins only
 - D) Fungal cell walls prevent support
14. Which statement about the cytoplasm is correct?
- A) Cytoplasm is the jelly-like material inside cells ✓
 - B) Cytoplasm stores hereditary information only
 - C) Cytoplasm exists outside the cell
 - D) Cytoplasm contains no organelles
15. Which statement correctly describes cytoplasmic functions?
- A) Many metabolic reactions occur in the cytoplasm ✓
 - B) Cytoplasm prevents enzyme activity
 - C) Cytoplasm contains no water
 - D) Cytoplasm blocks transport completely
16. Which statement about cytosol is correct?
- A) Cytosol is the fluid part of cytoplasm ✓
 - B) Cytosol is a membrane-bound organelle
 - C) Cytosol contains no dissolved substances
 - D) Cytosol prevents metabolism
17. Which statement correctly explains the role of cytoplasm in transport?
- A) Cytoplasm helps distribute materials within the cell ✓
 - B) Cytoplasm blocks intracellular movement
 - C) Cytoplasm stores oxygen only
 - D) Cytoplasm prevents diffusion
18. Which statement about organelles in cytoplasm is correct?
- A) Organelles are suspended in the cytoplasm ✓
 - B) Organelles exist outside the cell membrane
 - C) Organelles prevent metabolism
 - D) Organelles contain no enzymes
19. Which statement correctly explains diffusion across membranes?
- A) Diffusion moves substances from high to low concentration ✓
 - B) Diffusion requires ATP always
 - C) Diffusion prevents transport
 - D) Diffusion occurs only in solids
20. Which statement about osmosis is correct?
- A) Osmosis is movement of water across a membrane ✓
 - B) Osmosis transports proteins only
 - C) Osmosis prevents water balance
 - D) Osmosis occurs without membranes
21. Which statement correctly describes passive transport?
- A) Passive transport does not require cellular energy ✓

- B) Passive transport always uses ATP
 - C) Passive transport occurs against gradients only
 - D) Passive transport blocks membrane movement
22. Which statement about active transport is correct?
- A) Active transport requires energy to move substances ✓
 - B) Active transport occurs without ATP
 - C) Active transport always follows concentration gradients
 - D) Active transport prevents nutrient uptake
23. Which statement correctly explains facilitated diffusion?
- A) Facilitated diffusion uses transport proteins ✓
 - B) Facilitated diffusion requires ATP always
 - C) Facilitated diffusion blocks membrane channels
 - D) Facilitated diffusion occurs only in dead cells
24. Which statement about channel proteins is correct?
- A) Channel proteins allow specific ions or molecules to pass ✓
 - B) Channel proteins digest carbohydrates
 - C) Channel proteins prevent transport
 - D) Channel proteins are carbohydrates
25. Which statement correctly describes carrier proteins?
- A) Carrier proteins transport molecules across membranes ✓
 - B) Carrier proteins contain no amino acids
 - C) Carrier proteins stop osmosis
 - D) Carrier proteins are lipids only
26. Which statement about endocytosis is correct?
- A) Endocytosis allows cells to take in materials ✓
 - B) Endocytosis releases materials outside the cell
 - C) Endocytosis occurs without membranes
 - D) Endocytosis prevents vesicle formation
27. Which statement correctly explains phagocytosis?
- A) Phagocytosis involves engulfing large particles ✓
 - B) Phagocytosis releases proteins only
 - C) Phagocytosis prevents nutrient intake
 - D) Phagocytosis occurs only in plants
28. Which statement about pinocytosis is correct?
- A) Pinocytosis allows uptake of liquids ✓
 - B) Pinocytosis releases water only
 - C) Pinocytosis prevents diffusion
 - D) Pinocytosis occurs without vesicles
29. Which statement correctly describes exocytosis?
- A) Exocytosis exports substances outside the cell ✓
 - B) Exocytosis imports nutrients only

- C) Exocytosis prevents secretion
- D) Exocytosis occurs without energy

30. Which statement about membrane receptors is correct?

- A) Receptors receive specific chemical signals ✓
- B) Receptors digest lipids
- C) Receptors prevent communication
- D) Receptors store DNA

31. Which statement correctly explains membrane transport proteins?

- A) Transport proteins help move substances across membranes ✓
- B) Transport proteins prevent all movement
- C) Transport proteins are carbohydrates only
- D) Transport proteins contain no amino acids

32. Which statement about concentration gradients is correct?

- A) Substances move down concentration gradients during diffusion ✓
- B) Concentration gradients prevent transport
- C) Diffusion moves substances from low to high concentration only
- D) Concentration gradients contain ATP

33. Which statement correctly describes isotonic solutions?

- A) Water movement is balanced across the membrane ✓
- B) Cells burst immediately in isotonic solutions
- C) Isotonic solutions contain no water
- D) Isotonic conditions prevent osmosis

34. Which statement about hypotonic solutions is correct?

- A) Water enters the cell in hypotonic solutions ✓
- B) Water leaves the cell completely
- C) Hypotonic solutions prevent osmosis
- D) Hypotonic solutions contain no solutes

35. Which statement correctly explains hypertonic solutions?

- A) Water leaves the cell in hypertonic conditions ✓
- B) Hypertonic solutions force water into cells
- C) Hypertonic solutions contain pure water only
- D) Hypertonic solutions prevent membrane transport

36. Which statement about membrane surface area is correct?

- A) Larger membrane surface area improves exchange efficiency ✓
- B) Surface area prevents diffusion
- C) Surface area has no effect on transport
- D) Larger surface area blocks osmosis

37. Which statement correctly describes membrane permeability?

- A) Small nonpolar molecules cross membranes easily ✓
- B) All molecules cross membranes equally
- C) Membranes are impermeable to all substances
- D) Large molecules always diffuse freely

38. Which statement about ATP in active transport is correct?

- A) ATP provides energy for active transport ✓
- B) ATP prevents ion movement
- C) ATP is a membrane protein
- D) ATP blocks concentration gradients

39. Which statement correctly explains sodium-potassium pumps?

- A) They transport sodium and potassium ions using ATP ✓
- B) They perform diffusion only
- C) They prevent ion movement
- D) They are carbohydrates

40. Which statement about vesicles is correct?

- A) Vesicles help transport substances during endocytosis and exocytosis ✓
- B) Vesicles contain no membranes
- C) Vesicles prevent secretion
- D) Vesicles store DNA only

41. Which statement correctly describes membrane communication?

- A) Cell membranes help cells communicate with surroundings ✓
- B) Cell membranes prevent signaling
- C) Cell membranes store genetic information only
- D) Cell membranes block all interactions

42. Which statement about membrane flexibility is correct?

- A) Membrane flexibility allows cells to change shape ✓
- B) Membranes are completely rigid
- C) Membrane flexibility prevents transport
- D) Membrane flexibility destroys proteins

43. Which statement correctly explains the protective role of cell membranes?

- A) Cell membranes protect internal cellular contents ✓
- B) Cell membranes expose organelles directly
- C) Cell membranes contain no barriers
- D) Cell membranes prevent homeostasis

44. Which statement about cytoplasmic streaming is correct?

- A) Cytoplasmic streaming helps move materials within cells ✓
- B) Cytoplasmic streaming blocks intracellular movement
- C) Cytoplasmic streaming prevents metabolism
- D) Cytoplasmic streaming stores DNA

45. Which statement correctly describes membrane carbohydrates?

- A) Membrane carbohydrates assist in recognition and signaling ✓
- B) Membrane carbohydrates store ATP only
- C) Membrane carbohydrates prevent communication
- D) Membrane carbohydrates are proteins only

46. Which statement about transport specificity is correct?

- A) Membrane proteins often transport specific substances ✓

- B) All transport proteins move every molecule equally
- C) Transport specificity prevents nutrient uptake
- D) Specificity occurs only in plant cells

47. Which statement correctly explains why membranes are important?

- A) Membranes regulate transport and maintain homeostasis ✓
- B) Membranes prevent all cellular activity
- C) Membranes contain no biological function
- D) Membranes replace cytoplasm completely

48. Which statement about cytoplasm composition is correct?

- A) Cytoplasm contains water, ions, and biomolecules ✓
- B) Cytoplasm contains only proteins
- C) Cytoplasm lacks enzymes
- D) Cytoplasm contains no dissolved substances

49. Which statement correctly describes membrane-bound vesicle transport?

- A) Vesicles transport large molecules across membranes ✓
- B) Vesicle transport prevents secretion
- C) Vesicle transport occurs without membranes
- D) Vesicle transport destroys ATP

50. Which statement correctly explains the overall importance of membranes and cytoplasm?

- A) Membranes regulate transport while cytoplasm supports metabolism ✓
- B) Membranes and cytoplasm have no cellular roles
- C) Cytoplasm stores hereditary information only
- D) Membranes prevent biochemical reactions

1. Which statement correctly describes endoplasmic reticulum (ER)?

- A) ER is a network of membranes involved in synthesis and transport ✓
- B) ER stores genetic information
- C) ER produces ATP directly
- D) ER is found only in prokaryotes

2. Which statement about rough endoplasmic reticulum (RER) is correct?

- A) RER has ribosomes attached to its surface ✓
- B) RER lacks membranes
- C) RER produces lipids only
- D) RER is involved in DNA replication

3. Which statement about smooth endoplasmic reticulum (SER) is correct?

- A) SER is involved in lipid synthesis ✓
- B) SER contains ribosomes
- C) SER stores DNA
- D) SER produces proteins only

4. Which statement correctly describes the function of ER?

- A) ER helps in intracellular transport ✓
- B) ER stores oxygen

- C) ER prevents metabolism
- D) ER destroys ATP

5. Which statement about centrioles is correct?

- A) Centrioles help in cell division ✓
- B) Centrioles store genetic material
- C) Centrioles produce proteins
- D) Centrioles are found in plant cell walls

6. Which statement correctly describes centrioles location?

- A) Centrioles are found in animal cells ✓
- B) Centrioles are found in plant cell walls
- C) Centrioles are present in vacuoles
- D) Centrioles are in mitochondria

7. Which statement about vacuoles is correct?

- A) Vacuoles store water, nutrients, and waste ✓
- B) Vacuoles produce ATP
- C) Vacuoles synthesize proteins
- D) Vacuoles are absent in all plant cells

8. Which statement correctly describes plant vacuoles?

- A) Plant vacuoles are large and maintain turgor pressure ✓
- B) Plant vacuoles store DNA only
- C) Plant vacuoles produce enzymes
- D) Plant vacuoles are absent in plants

9. Which statement about mitochondria is correct?

- A) Mitochondria are the site of ATP production ✓
- B) Mitochondria store genetic material only
- C) Mitochondria prevent respiration
- D) Mitochondria are found only in bacteria

10. Which statement correctly describes mitochondrial DNA?

- A) Mitochondria contain their own DNA ✓
- B) Mitochondria lack genetic material
- C) Mitochondrial DNA is found in nucleus only
- D) Mitochondrial DNA prevents respiration

11. Which statement about mitochondria function is correct?

- A) Mitochondria carry out cellular respiration ✓
- B) Mitochondria digest proteins only
- C) Mitochondria produce carbohydrates
- D) Mitochondria store water

12. Which statement correctly describes cilia?

- A) Cilia are short hair-like structures for movement ✓
- B) Cilia store DNA
- C) Cilia produce ATP
- D) Cilia are present only in plants

13. Which statement about flagella is correct?
A) Flagella are long structures used for cell movement ✓
B) Flagella store lipids
C) Flagella prevent movement
D) Flagella are found only in vacuoles
14. Which statement correctly describes difference between cilia and flagella?
A) Cilia are shorter and more numerous than flagella ✓
B) Flagella are shorter than cilia always
C) Both are non-motile structures
D) Cilia are found only in bacteria
15. Which statement about ribosomes is correct?
A) Ribosomes are sites of protein synthesis ✓
B) Ribosomes produce ATP
C) Ribosomes store genetic material
D) Ribosomes are membrane-bound organelles
16. Which statement correctly describes ribosome composition?
A) Ribosomes contain RNA and proteins ✓
B) Ribosomes contain lipids only
C) Ribosomes contain DNA only
D) Ribosomes are made of carbohydrates
17. Which statement about lysosomes is correct?
A) Lysosomes contain digestive enzymes ✓
B) Lysosomes produce ATP
C) Lysosomes store DNA
D) Lysosomes prevent digestion
18. Which statement correctly describes lysosome function?
A) Lysosomes break down waste materials ✓
B) Lysosomes synthesize proteins
C) Lysosomes store oxygen
D) Lysosomes produce lipids
19. Which statement about ER in protein synthesis is correct?
A) RER assists in protein synthesis and modification ✓
B) SER produces DNA
C) ER prevents translation
D) ER destroys ribosomes
20. Which statement correctly explains ER and Golgi interaction?
A) ER transports proteins to Golgi apparatus ✓
B) ER prevents secretion
C) ER stores waste permanently
D) ER replaces Golgi function
21. Which statement about centrioles role in mitosis is correct?
A) Centrioles organize spindle fibers ✓

- B) Centrioles produce ATP
- C) Centrioles store proteins
- D) Centrioles form cell walls

22. Which statement correctly describes vacuole function in plants?

- A) Vacuoles maintain cell turgidity ✓
- B) Vacuoles produce enzymes only
- C) Vacuoles prevent water uptake
- D) Vacuoles store DNA

23. Which statement about mitochondrial structure is correct?

- A) Mitochondria have inner folded membranes called cristae ✓
- B) Mitochondria lack membranes
- C) Mitochondria are flat structures
- D) Mitochondria contain cell walls

24. Which statement correctly describes ATP production?

- A) ATP is produced in mitochondria ✓
- B) ATP is produced in ribosomes only
- C) ATP is stored in vacuoles
- D) ATP is produced in cell wall

25. Which statement about cilia function is correct?

- A) Cilia help move substances across cell surfaces ✓
- B) Cilia store waste
- C) Cilia produce proteins
- D) Cilia prevent movement

26. Which statement correctly describes flagella in sperm cells?

- A) Flagella help sperm movement ✓
- B) Flagella store energy
- C) Flagella prevent fertilization
- D) Flagella are absent in sperm

27. Which statement about ribosomes in prokaryotes is correct?

- A) Prokaryotes have smaller ribosomes ✓
- B) Prokaryotes lack ribosomes
- C) Ribosomes exist only in nucleus
- D) Ribosomes are membrane-bound

28. Which statement correctly explains lysosomes in animal cells?

- A) Lysosomes help in intracellular digestion ✓
- B) Lysosomes produce DNA
- C) Lysosomes store water
- D) Lysosomes form cell walls

29. Which statement about SER detoxification is correct?

- A) SER helps detoxify harmful substances ✓
- B) SER produces ribosomes

- C) SER stores DNA
- D) SER prevents metabolism

30. Which statement correctly describes ER membrane continuity?

- A) ER is continuous with nuclear membrane ✓
- B) ER is separate from all organelles
- C) ER exists outside the cell
- D) ER has no membranes

31. Which statement about vacuoles in animal cells is correct?

- A) Animal vacuoles are smaller than plant vacuoles ✓
- B) Animal vacuoles are always absent
- C) Animal vacuoles produce ATP
- D) Animal vacuoles store DNA only

32. Which statement correctly describes mitochondrial role in metabolism?

- A) Mitochondria convert glucose into energy ✓
- B) Mitochondria store proteins
- C) Mitochondria prevent respiration
- D) Mitochondria digest nutrients

33. Which statement about cristae is correct?

- A) Cristae increase surface area for ATP production ✓
- B) Cristae prevent respiration
- C) Cristae store DNA
- D) Cristae are in ribosomes

34. Which statement correctly describes ribosomal function in cells?

- A) Ribosomes translate mRNA into proteins ✓
- B) Ribosomes store glucose
- C) Ribosomes prevent transcription
- D) Ribosomes produce lipids

35. Which statement about lysosomes and autophagy is correct?

- A) Lysosomes break down damaged organelles ✓
- B) Lysosomes produce energy
- C) Lysosomes synthesize proteins
- D) Lysosomes form DNA

36. Which statement correctly explains ciliary movement?

- A) Cilia move in coordinated waves ✓
- B) Cilia are rigid structures
- C) Cilia prevent transport
- D) Cilia store nutrients

37. Which statement about flagella structure is correct?

- A) Flagella contain microtubules ✓
- B) Flagella contain cellulose
- C) Flagella are lipid structures only
- D) Flagella lack proteins

38. Which statement correctly describes ER types?

- A) ER is divided into rough and smooth types ✓
- B) ER has only one form
- C) ER is only found in bacteria
- D) ER has no functions

39. Which statement about centrioles duplication is correct?

- A) Centrioles duplicate before cell division ✓
- B) Centrioles never duplicate
- C) Centrioles produce DNA
- D) Centrioles form membranes

40. Which statement correctly describes vacuole membrane?

- A) Vacuole is surrounded by tonoplast ✓
- B) Vacuole has no membrane
- C) Vacuole is made of DNA
- D) Vacuole is a lipid only

41. Which statement about mitochondrial inheritance is correct?

- A) Mitochondria are maternally inherited in many organisms ✓
- B) Mitochondria are paternally inherited only
- C) Mitochondria have no inheritance
- D) Mitochondria are formed daily from ribosomes

42. Which statement correctly describes ribosomes location?

- A) Ribosomes can be free or attached to ER ✓
- B) Ribosomes are only in nucleus
- C) Ribosomes are only in vacuoles
- D) Ribosomes are only in cell wall

43. Which statement about lysosomal enzymes is correct?

- A) Lysosomal enzymes function in acidic conditions ✓
- B) Lysosomal enzymes work only in neutral pH
- C) Lysosomal enzymes are DNA molecules
- D) Lysosomal enzymes produce ATP

44. Which statement correctly describes ER function in lipids?

- A) SER is involved in lipid synthesis ✓
- B) RER synthesizes lipids only
- C) ER prevents lipid formation
- D) ER stores glycogen

45. Which statement about cell motility is correct?

- A) Cilia and flagella are involved in movement ✓
- B) Cilia prevent movement
- C) Flagella store nutrients
- D) Both are non-functional structures

46. Which statement correctly describes mitochondria as semi-autonomous organelles?

- A) They contain their own DNA and ribosomes ✓

- B) They lack all genetic material
- C) They depend completely on nucleus
- D) They are not living structures

47. Which statement about vacuole function in storage is correct?

- A) Vacuoles store nutrients and waste materials ✓
- B) Vacuoles produce proteins
- C) Vacuoles synthesize DNA
- D) Vacuoles prevent osmosis

48. Which statement correctly explains ribosomes in protein synthesis sites?

- A) Ribosomes assemble amino acids into proteins ✓
- B) Ribosomes break down proteins only
- C) Ribosomes store fats
- D) Ribosomes prevent translation

49. Which statement about lysosomes in cell death is correct?

- A) Lysosomes can contribute to programmed cell death ✓
- B) Lysosomes prevent all cell death
- C) Lysosomes produce DNA
- D) Lysosomes store oxygen

50. Which statement correctly summarizes organelle functions?

- A) Organelles work together to maintain cell function ✓
- B) Organelles act independently without interaction
- C) Organelles have no role in metabolism
- D) Organelles exist only in plants

1. Which statement correctly describes ATP?

- A) ATP is the main energy currency of the cell ✓
- B) ATP stores genetic information
- C) ATP is a structural lipid
- D) ATP is a carbohydrate polymer

2. Which statement about ATP structure is correct?

- A) ATP consists of adenine, ribose, and three phosphate groups ✓
- B) ATP contains amino acids
- C) ATP is made of fatty acids
- D) ATP lacks phosphate groups

3. Which statement correctly describes ATP hydrolysis?

- A) ATP hydrolysis releases energy by breaking phosphate bonds ✓
- B) ATP hydrolysis stores energy
- C) ATP hydrolysis prevents reactions
- D) ATP hydrolysis produces DNA

4. Which statement about enzymes is correct?

- A) Enzymes are biological catalysts that speed up reactions ✓
- B) Enzymes are carbohydrates only

- C) Enzymes destroy all substrates
D) Enzymes store genetic material
5. Which statement correctly describes activation energy?
A) Activation energy is the energy needed to start a reaction ✓
B) Activation energy is the final energy of products
C) Activation energy prevents reactions permanently
D) Activation energy is ATP itself
6. Which statement about enzyme function is correct?
A) Enzymes lower activation energy ✓
B) Enzymes increase activation energy
C) Enzymes are consumed in reactions
D) Enzymes stop all metabolism
7. Which statement correctly describes exergonic reactions?
A) Exergonic reactions release energy ✓
B) Exergonic reactions require energy input only
C) Exergonic reactions store ATP permanently
D) Exergonic reactions are non-spontaneous only
8. Which statement about endergonic reactions is correct?
A) Endergonic reactions require energy input ✓
B) Endergonic reactions release energy always
C) Endergonic reactions destroy ATP
D) Endergonic reactions are always spontaneous
9. Which statement correctly explains energy coupling?
A) Energy from ATP drives endergonic reactions ✓
B) Energy coupling prevents ATP use
C) Energy coupling destroys enzymes
D) Energy coupling occurs only in water
10. Which statement about ATP synthesis is correct?
A) ATP is formed from ADP and inorganic phosphate ✓
B) ATP is formed from proteins
C) ATP is made from lipids only
D) ATP is not regenerated
11. Which statement correctly describes ADP?
A) ADP is adenosine diphosphate ✓
B) ADP is a protein
C) ADP is a lipid
D) ADP contains no phosphate
12. Which statement about phosphorylation is correct?
A) Phosphorylation adds a phosphate group to a molecule ✓
B) Phosphorylation removes all energy
C) Phosphorylation destroys enzymes
D) Phosphorylation stops ATP formation

13. Which statement correctly describes enzyme specificity?
- A) Enzymes act on specific substrates ✓
 - B) Enzymes act on all molecules equally
 - C) Enzymes are non-selective always
 - D) Enzymes do not bind substrates
14. Which statement about active site is correct?
- A) Active site is where substrate binds ✓
 - B) Active site is DNA region
 - C) Active site produces ATP
 - D) Active site is outside enzyme
15. Which statement correctly explains enzyme-substrate complex?
- A) It is a temporary complex formed during reaction ✓
 - B) It is permanent and irreversible
 - C) It prevents product formation
 - D) It destroys enzymes
16. Which statement about enzyme inhibition is correct?
- A) Inhibitors reduce enzyme activity ✓
 - B) Inhibitors increase all reactions
 - C) Inhibitors are always substrates
 - D) Inhibitors produce ATP
17. Which statement correctly describes competitive inhibition?
- A) Inhibitor competes with substrate for active site ✓
 - B) Inhibitor binds DNA
 - C) Inhibitor destroys enzyme permanently
 - D) Inhibitor increases enzyme speed
18. Which statement about non-competitive inhibition is correct?
- A) Inhibitor binds to allosteric site ✓
 - B) Inhibitor binds active site only
 - C) Inhibitor increases substrate binding
 - D) Inhibitor prevents all proteins
19. Which statement correctly explains allosteric regulation?
- A) Enzyme activity is controlled by binding at regulatory site ✓
 - B) Enzymes cannot change shape
 - C) Allosteric regulation destroys enzymes
 - D) It occurs only in DNA
20. Which statement about feedback inhibition is correct?
- A) End product inhibits earlier enzyme steps ✓
 - B) Feedback increases all reactions
 - C) Feedback prevents ATP production completely
 - D) Feedback occurs only in plants
21. Which statement correctly describes enzyme denaturation?
- A) Denaturation changes enzyme shape and function ✓

- B) Denaturation increases activity always
- C) Denaturation forms ATP
- D) Denaturation is reversible always

22. Which statement about temperature effect on enzymes is correct?

- A) High temperature can denature enzymes ✓
- B) Temperature never affects enzymes
- C) Enzymes work best at all temperatures
- D) Cold destroys all enzymes permanently

23. Which statement correctly explains optimum temperature?

- A) Enzymes work best at a specific temperature ✓
- B) Enzymes work best at any temperature
- C) Enzymes stop working at optimum temperature
- D) Optimum temperature destroys enzymes

24. Which statement about pH effect is correct?

- A) Enzymes function best at specific pH ✓
- B) pH has no effect on enzymes
- C) All enzymes work at pH 14
- D) pH destroys ATP always

25. Which statement correctly describes enzyme cofactors?

- A) Cofactors assist enzyme activity ✓
- B) Cofactors destroy substrates
- C) Cofactors are proteins only
- D) Cofactors prevent catalysis

26. Which statement about coenzymes is correct?

- A) Coenzymes are organic cofactors ✓
- B) Coenzymes are DNA molecules
- C) Coenzymes prevent reactions
- D) Coenzymes are lipids only

27. Which statement correctly explains vitamins in enzyme function?

- A) Some vitamins act as coenzymes ✓
- B) Vitamins are enzymes themselves
- C) Vitamins destroy ATP
- D) Vitamins prevent metabolism

28. Which statement about energy in reactions is correct?

- A) Energy is required or released depending on reaction type ✓
- B) All reactions release energy
- C) No reaction involves energy
- D) Energy is unrelated to chemistry

29. Which statement correctly describes spontaneous reactions?

- A) Spontaneous reactions occur without continuous energy input ✓
- B) Spontaneous reactions always need ATP

- C) Spontaneous reactions are impossible
D) Spontaneous reactions store energy only
30. Which statement about Gibbs free energy is correct?
A) It determines whether a reaction is spontaneous ✓
B) It measures enzyme size
C) It destroys ATP
D) It is unrelated to reactions
31. Which statement correctly explains negative ΔG ?
A) Negative ΔG indicates exergonic reaction ✓
B) Negative ΔG indicates endergonic reaction
C) Negative ΔG means no reaction occurs
D) Negative ΔG destroys enzymes
32. Which statement about positive ΔG is correct?
A) Positive ΔG indicates endergonic reaction ✓
B) Positive ΔG releases energy
C) Positive ΔG means ATP is not needed
D) Positive ΔG prevents metabolism
33. Which statement correctly describes ATP role in metabolism?
A) ATP transfers energy between reactions ✓
B) ATP stores DNA
C) ATP is a structural protein
D) ATP prevents energy flow
34. Which statement about ATP regeneration is correct?
A) ATP is continuously regenerated in cells ✓
B) ATP is used once only
C) ATP cannot be reused
D) ATP is permanently stored
35. Which statement correctly explains energy currency?
A) ATP is called energy currency because it transfers energy ✓
B) ATP stores genes
C) ATP is a lipid
D) ATP prevents respiration
36. Which statement about enzyme shape is correct?
A) Enzyme shape determines its function ✓
B) Shape has no effect on enzymes
C) Enzymes do not fold
D) Enzymes are always rigid
37. Which statement correctly describes induced fit model?
A) Enzyme changes shape slightly when substrate binds ✓
B) Enzyme is completely rigid
C) Substrate changes DNA
D) Enzyme is destroyed after reaction

38. Which statement about lock and key model is correct?

- A) Substrate fits exactly into active site ✓
- B) Enzyme has no active site
- C) Substrate changes enzyme permanently
- D) Enzymes are non-specific

39. Which statement correctly explains reaction rate?

- A) Enzymes increase reaction rate ✓
- B) Enzymes decrease all reactions
- C) Reaction rate is unrelated to enzymes
- D) ATP stops reactions

40. Which statement about enzyme concentration is correct?

- A) Increasing enzyme concentration increases reaction rate ✓
- B) More enzymes slow reactions
- C) Enzymes have no effect
- D) Enzymes destroy substrates always

41. Which statement correctly describes substrate concentration effect?

- A) Reaction rate increases until enzymes are saturated ✓
- B) Substrate has no effect
- C) Substrate destroys enzymes
- D) Substrate stops reactions

42. Which statement about enzyme reuse is correct?

- A) Enzymes are not consumed in reactions ✓
- B) Enzymes are used once
- C) Enzymes are destroyed after reaction
- D) Enzymes become substrates

43. Which statement correctly explains metabolic pathways?

- A) Enzymes regulate steps in metabolic pathways ✓
- B) Pathways occur without enzymes
- C) Pathways destroy ATP
- D) Pathways are random reactions

44. Which statement about coupling ATP to reactions is correct?

- A) ATP hydrolysis drives unfavorable reactions ✓
- B) ATP prevents all reactions
- C) ATP is not used in cells
- D) ATP blocks metabolism

45. Which statement correctly describes exergonic vs endergonic relationship?

- A) Exergonic reactions can drive endergonic ones ✓
- B) Both always release energy
- C) Both always require no energy
- D) They are unrelated

46. Which statement about enzyme regulation is correct?

- A) Enzyme activity is tightly controlled in cells ✓

- B) Enzymes are never regulated
- C) Regulation destroys enzymes
- D) Regulation prevents metabolism

47. Which statement correctly explains metabolic control?

- A) Cells regulate enzymes to maintain balance ✓
- B) Cells do not control enzymes
- C) Metabolism is random
- D) ATP has no role

48. Which statement about enzyme localization is correct?

- A) Some enzymes are located in specific organelles ✓
- B) Enzymes float randomly without function
- C) Enzymes exist only outside cells
- D) Enzymes are not organized

49. Which statement correctly describes energy transformation in cells?

- A) Cells convert chemical energy into usable ATP ✓
- B) Cells destroy all energy
- C) Energy cannot change form
- D) ATP prevents energy use

50. Which statement correctly summarizes enzymes and ATP?

- A) Enzymes control reactions and ATP provides energy ✓
- B) ATP controls enzymes directly only
- C) Enzymes store energy permanently
- D) Both are unrelated to metabolism

1. Which statement correctly describes DNA?

- A) DNA stores genetic information ✓
- B) DNA is a lipid molecule
- C) DNA is a carbohydrate
- D) DNA is an enzyme

2. Which statement about the nuclear envelope is correct?

- A) It surrounds and protects the nucleus ✓
- B) It produces ATP
- C) It synthesizes proteins
- D) It stores lipids

3. Which statement correctly describes the structure of nuclear envelope?

- A) It is a double membrane structure ✓
- B) It is a single protein layer only
- C) It is made of cellulose
- D) It contains ribosomes only

4. Which statement about nuclear pores is correct?

- A) Nuclear pores regulate transport between nucleus and cytoplasm ✓
- B) Nuclear pores store DNA

- C) Nuclear pores prevent all movement
D) Nuclear pores synthesize proteins
5. Which statement correctly describes DNA discovery?
A) DNA was identified as genetic material through multiple experiments ✓
B) DNA was discovered as a lipid
C) DNA was first observed in mitochondria only
D) DNA was discovered as a carbohydrate
6. Which statement about Frederick Miescher is correct?
A) He first isolated DNA from nuclei ✓
B) He discovered proteins only
C) He discovered ribosomes
D) He proved RNA structure
7. Which statement correctly describes Griffith's experiment?
A) It showed bacterial transformation ✓
B) It proved protein is genetic material
C) It discovered ATP
D) It showed lipid inheritance
8. Which statement about Avery, MacLeod, and McCarty is correct?
A) They proved DNA is genetic material ✓
B) They discovered ribosomes
C) They discovered mitochondria
D) They proved RNA is enzyme
9. Which statement correctly describes Hershey-Chase experiment?
A) It confirmed DNA is genetic material ✓
B) It showed proteins carry inheritance
C) It discovered cell membrane
D) It described enzyme function
10. Which statement about DNA structure is correct?
A) DNA is a double-stranded molecule ✓
B) DNA is single-stranded only
C) DNA contains no bases
D) DNA is a lipid polymer
11. Which statement correctly describes nucleotides?
A) Nucleotides are building blocks of DNA ✓
B) Nucleotides are amino acids
C) Nucleotides are lipids
D) Nucleotides contain no phosphate
12. Which statement about DNA sugar is correct?
A) DNA contains deoxyribose sugar ✓
B) DNA contains ribose sugar only
C) DNA contains glucose
D) DNA contains fructose

13. Which statement correctly describes nitrogenous bases?
- A) DNA contains adenine, thymine, guanine, cytosine ✓
 - B) DNA contains uracil
 - C) DNA contains only amino acids
 - D) DNA contains lipids only
14. Which statement about base pairing is correct?
- A) Adenine pairs with thymine ✓
 - B) Adenine pairs with cytosine
 - C) Guanine pairs with thymine
 - D) Cytosine pairs with adenine only
15. Which statement correctly describes guanine-cytosine pairing?
- A) Guanine pairs with cytosine ✓
 - B) Guanine pairs with adenine
 - C) Cytosine pairs with thymine
 - D) Guanine does not pair
16. Which statement about hydrogen bonds in DNA is correct?
- A) Hydrogen bonds hold two DNA strands together ✓
 - B) Hydrogen bonds form peptide bonds
 - C) Hydrogen bonds destroy DNA
 - D) Hydrogen bonds store ATP
17. Which statement correctly describes DNA backbone?
- A) DNA backbone consists of sugar and phosphate ✓
 - B) DNA backbone consists of proteins
 - C) DNA backbone consists of lipids
 - D) DNA backbone contains nitrogen bases only
18. Which statement about antiparallel DNA strands is correct?
- A) DNA strands run in opposite directions ✓
 - B) DNA strands run in same direction
 - C) DNA strands are parallel always
 - D) DNA strands are random
19. Which statement correctly describes Watson and Crick model?
- A) They proposed the double helix structure ✓
 - B) They discovered ribosomes
 - C) They discovered proteins
 - D) They proved enzyme function
20. Which statement about Rosalind Franklin is correct?
- A) She contributed X-ray diffraction data for DNA ✓
 - B) She discovered ATP
 - C) She discovered mitochondria
 - D) She proved lipid structure
21. Which statement correctly describes double helix?
- A) DNA twists like a spiral ladder ✓

- B) DNA is a straight chain
- C) DNA is circular only
- D) DNA has no structure

22. Which statement about DNA replication is correct?

- A) DNA replication produces identical copies ✓
- B) DNA replication destroys genetic material
- C) DNA replication occurs only in RNA
- D) DNA replication prevents cell division

23. Which statement correctly describes semi-conservative replication?

- A) Each new DNA has one old and one new strand ✓
- B) DNA is completely new each time
- C) DNA is destroyed during replication
- D) DNA is copied without strands

24. Which statement about histones is correct?

- A) Histones are proteins that package DNA ✓
- B) Histones are lipids
- C) Histones are carbohydrates
- D) Histones destroy DNA

25. Which statement correctly describes chromatin?

- A) Chromatin is DNA-protein complex ✓
- B) Chromatin is pure DNA only
- C) Chromatin is RNA only
- D) Chromatin is lipid material

26. Which statement about chromosomes is correct?

- A) Chromosomes are condensed chromatin ✓
- B) Chromosomes are proteins only
- C) Chromosomes are found in cytoplasm only
- D) Chromosomes lack DNA

27. Which statement correctly describes nucleosomes?

- A) DNA wrapped around histone proteins ✓
- B) DNA wrapped around lipids
- C) RNA wrapped around proteins
- D) Proteins wrapped around ATP

28. Which statement about DNA packaging is correct?

- A) DNA is highly compacted to fit in nucleus ✓
- B) DNA is loosely floating without organization
- C) DNA is stored in cytoplasm only
- D) DNA does not require packaging

29. Which statement correctly describes higher-order DNA folding?

- A) Chromatin forms loops and coils ✓
- B) DNA remains straight always

- C) DNA breaks during folding
- D) DNA loses function when folded

30. Which statement about euchromatin is correct?

- A) Euchromatin is loosely packed and active ✓
- B) Euchromatin is inactive always
- C) Euchromatin is protein only
- D) Euchromatin is found outside nucleus

31. Which statement correctly describes heterochromatin?

- A) Heterochromatin is tightly packed and less active ✓
- B) Heterochromatin is fully active
- C) Heterochromatin is RNA only
- D) Heterochromatin is absent in cells

32. Which statement about DNA function is correct?

- A) DNA controls inheritance and protein synthesis ✓
- B) DNA stores energy only
- C) DNA produces lipids
- D) DNA is not functional

33. Which statement correctly describes gene location?

- A) Genes are located on chromosomes ✓
- B) Genes are in cytoplasm only
- C) Genes are in mitochondria only
- D) Genes are outside cells

34. Which statement about transcription is correct?

- A) Transcription produces RNA from DNA ✓
- B) Transcription produces proteins directly
- C) Transcription destroys DNA
- D) Transcription occurs in cytoplasm only

35. Which statement correctly describes DNA stability?

- A) DNA is stable due to base pairing ✓
- B) DNA is unstable always
- C) DNA breaks easily in normal conditions
- D) DNA has no structure

36. Which statement about replication enzymes is correct?

- A) DNA polymerase helps in DNA replication ✓
- B) DNA polymerase destroys DNA
- C) DNA polymerase is a lipid
- D) DNA polymerase stores ATP

37. Which statement correctly describes helicase function?

- A) Helicase unwinds DNA strands ✓
- B) Helicase joins amino acids
- C) Helicase forms membranes
- D) Helicase stores DNA

38. Which statement about ligase is correct?
- A) Ligase joins DNA fragments ✓
 - B) Ligase breaks DNA permanently
 - C) Ligase is a carbohydrate
 - D) Ligase prevents replication
39. Which statement correctly describes replication fork?
- A) It is the site where DNA unwinds ✓
 - B) It is a protein structure only
 - C) It stores genetic information
 - D) It prevents DNA synthesis
40. Which statement about DNA mutations is correct?
- A) Mutations are changes in DNA sequence ✓
 - B) Mutations always destroy cells
 - C) Mutations prevent inheritance
 - D) Mutations occur only in proteins
41. Which statement correctly describes DNA repair?
- A) Cells have mechanisms to fix DNA damage ✓
 - B) DNA cannot be repaired
 - C) Repair destroys genes
 - D) Repair prevents replication
42. Which statement about chromosome number is correct?
- A) Species have specific chromosome numbers ✓
 - B) All organisms have same chromosomes
 - C) Chromosomes are not inherited
 - D) Chromosomes are identical in all species
43. Which statement correctly describes mitochondrial DNA?
- A) Mitochondria contain small circular DNA ✓
 - B) Mitochondria lack DNA
 - C) Mitochondrial DNA is nuclear only
 - D) Mitochondrial DNA is protein based
44. Which statement about DNA function in nucleus is correct?
- A) DNA is stored and protected in nucleus ✓
 - B) DNA is stored in cytoplasm only
 - C) DNA is outside cell membrane
 - D) DNA is not protected
45. Which statement correctly describes chromosomal proteins?
- A) Proteins help organize and regulate DNA ✓
 - B) Proteins destroy DNA
 - C) Proteins are DNA itself
 - D) Proteins prevent replication
46. Which statement about DNA length is correct?
- A) DNA is very long and highly compacted ✓

- B) DNA is very short always
- C) DNA has no measurable length
- D) DNA is single amino acid

47. Which statement correctly describes genome?

- A) Genome is the complete set of DNA in a cell ✓
- B) Genome is a single protein
- C) Genome is a lipid layer
- D) Genome is only RNA

48. Which statement about DNA function in protein synthesis is correct?

- A) DNA provides instructions for protein synthesis ✓
- B) DNA directly becomes protein
- C) DNA destroys ribosomes
- D) DNA prevents translation

49. Which statement correctly describes nuclear organization?

- A) DNA is organized into chromosomes inside nucleus ✓
- B) DNA floats freely in cytoplasm
- C) DNA has no structure in nucleus
- D) DNA is outside cells always

50. Which statement correctly summarizes DNA and nuclear structure?

- A) DNA is packed into chromosomes inside nucleus for storage and function ✓
- B) DNA is unorganized and free in cells
- C) DNA exists only in membranes
- D) DNA has no relation to nucleus

1. RNA differs from DNA mainly because it contains:

- A) Thymine
- B) Deoxyribose
- C) Uracil ✓
- D) Double strands

2. The sugar present in RNA is:

- A) Ribose ✓
- B) Deoxyribose
- C) Glucose
- D) Fructose

3. RNA is usually:

- A) Double-stranded
- B) Single-stranded ✓

- C) Triple-stranded
 - D) Circular always
-

4. DNA replication is best described as:

- A) Conservative
 - B) Semiconservative ✓
 - C) Dispersive only
 - D) Random
-

5. The enzyme that unwinds DNA is:

- A) Ligase
 - B) Polymerase
 - C) Helicase ✓
 - D) Primase
-

6. DNA helicase function is to:

- A) Join fragments
 - B) Break hydrogen bonds ✓
 - C) Add nucleotides
 - D) Remove primers
-

7. The replication fork is:

- A) DNA end point
 - B) Site where DNA unwinds ✓
 - C) RNA structure
 - D) Protein complex
-

8. DNA polymerase synthesizes DNA in:

- A) 3' to 5' direction
 - B) 5' to 3' direction ✓
 - C) Both directions
 - D) Random direction
-

9. DNA polymerase requires:

- A) Lipids
- B) RNA primer ✓
- C) ATP only
- D) Ribosomes

10. RNA primer is synthesized by:

- A) DNA polymerase
 - B) Primase ✓
 - C) Helicase
 - D) Ligase
-

11. RNA primer is important because it:

- A) Stops replication
 - B) Provides free 3'-OH group ✓
 - C) Breaks DNA
 - D) Forms proteins
-

12. Okazaki fragments are found in:

- A) Leading strand
 - B) Lagging strand ✓
 - C) RNA only
 - D) Mitochondria only
-

13. The leading strand is synthesized:

- A) Discontinuously
 - B) Continuously ✓
 - C) Randomly
 - D) Backward
-

14. The lagging strand is synthesized:

- A) Continuously
 - B) Discontinuously ✓
 - C) Not at all
 - D) Only in nucleus
-

15. DNA ligase function is to:

- A) Break DNA
 - B) Join Okazaki fragments ✓
 - C) Unwind DNA
 - D) Make RNA
-

16. RNA contains which base instead of thymine?

- A) Adenine
 - B) Uracil ✓
 - C) Cytosine
 - D) Guanine
-

17. DNA replication occurs during:

- A) G1 phase
 - B) S phase ✓
 - C) M phase
 - D) G0 phase
-

18. The replication fork moves due to:

- A) DNA polymerase
 - B) Helicase activity ✓
 - C) Ribosomes
 - D) tRNA
-

19. DNA polymerase adds nucleotides at:

- A) 5' end
 - B) 3' end ✓
 - C) Both ends
 - D) Random sites
-

20. RNA primer is later removed by:

- A) Helicase
 - B) DNA polymerase I ✓
 - C) Ligase
 - D) Primase
-

21. DNA replication is said to be bidirectional because:

- A) Only one strand is copied
 - B) Two replication forks form ✓
 - C) RNA is used
 - D) Proteins are involved
-

22. The enzyme that stabilizes unwound DNA is:

- A) Helicase
- B) SSB proteins ✓

- C) Ligase
 - D) Primase
-

23. DNA strands are antiparallel because:

- A) Same direction
 - B) Opposite polarity ✓
 - C) Circular structure
 - D) Protein binding
-

24. RNA is mainly involved in:

- A) Storage of genetic material
 - B) Protein synthesis ✓
 - C) DNA repair only
 - D) Cell wall formation
-

25. DNA replication starts at:

- A) Telomere only
 - B) Origin of replication ✓
 - C) Centromere
 - D) Ribosome
-

26. DNA helicase uses energy from:

- A) Lipids
 - B) ATP ✓
 - C) Glucose
 - D) DNA
-

27. DNA polymerase is an example of:

- A) Lipase
 - B) Nuclease
 - C) Enzyme ✓
 - D) Hormone
-

28. RNA primer is made of:

- A) DNA nucleotides
- B) RNA nucleotides ✓
- C) Amino acids
- D) Lipids

29. The replication fork resembles a:

- A) Circle
 - B) Y-shape ✓
 - C) Line
 - D) Triangle
-

30. DNA replication ensures:

- A) Protein synthesis
 - B) Genetic continuity ✓
 - C) Cell death
 - D) Mutation always
-

31. The lagging strand is synthesized in fragments because:

- A) DNA is circular
 - B) Polymerase works 5'→3' only ✓
 - C) RNA is absent
 - D) Ligase is inactive
-

32. RNA is synthesized from DNA by:

- A) DNA polymerase
 - B) RNA polymerase ✓
 - C) Ligase
 - D) Helicase
-

33. DNA polymerase cannot start synthesis without:

- A) ATP
 - B) Primer ✓
 - C) RNA
 - D) Protein
-

34. Helicase breaks:

- A) Phosphodiester bonds
 - B) Hydrogen bonds ✓
 - C) Peptide bonds
 - D) Ionic bonds
-

35. Okazaki fragments are joined by:

- A) Primase
 - B) Ligase ✓
 - C) Helicase
 - D) RNA polymerase
-

36. DNA replication is essential for:

- A) Respiration
 - B) Cell division ✓
 - C) Digestion
 - D) Movement
-

37. RNA is usually found in:

- A) Nucleus only
 - B) Nucleus and cytoplasm ✓
 - C) Mitochondria only
 - D) Cell wall
-

38. DNA replication is highly accurate due to:

- A) Random enzymes
 - B) Proofreading by polymerase ✓
 - C) RNA primers
 - D) Lipids
-

39. DNA helicase acts at:

- A) Replication fork ✓
 - B) Ribosome
 - C) Cytoplasm
 - D) Golgi
-

40. RNA primer is removed during:

- A) Initiation
 - B) Elongation/processing ✓
 - C) Translation
 - D) Mutation
-

41. DNA replication is:

- A) One-way only
- B) Semi-bidirectional ✓

- C) Non-directional
 - D) Random
-

42. DNA polymerase works by adding nucleotides via:

- A) Peptide bonds
 - B) Phosphodiester bonds ✓
 - C) Hydrogen bonds
 - D) Ionic bonds
-

43. The main function of RNA in cells is:

- A) Genetic storage
 - B) Protein synthesis support ✓
 - C) DNA replication
 - D) Energy storage
-

44. DNA helicase is essential for:

- A) Protein folding
 - B) Strand separation ✓
 - C) RNA synthesis
 - D) Lipid formation
-

45. Replication fork formation requires:

- A) RNA polymerase
 - B) Helicase action ✓
 - C) Ribosome
 - D) tRNA
-

46. DNA replication occurs in:

- A) Cytoplasm only
 - B) Nucleus (eukaryotes) ✓
 - C) Ribosome
 - D) Golgi body
-

47. RNA primer is required for:

- A) Translation
- B) DNA synthesis initiation ✓
- C) Protein folding
- D) Cell division

48. DNA polymerase cannot synthesize in:

- A) 5'→3' direction
 - B) 3'→5' direction ✓
 - C) Both directions
 - D) Circular direction
-

49. The replication fork is stabilized by:

- A) SSB proteins ✓
 - B) Ligase
 - C) RNA primer
 - D) ATP
-

50. DNA replication ensures each daughter cell receives:

- A) Half proteins
 - B) Identical genetic material ✓
 - C) Random DNA
 - D) RNA only
-

50 MCQs

1. Topoisomerase functions during DNA replication by:

- A) Synthesizing RNA primer
 - B) Relieving supercoiling ahead of replication fork ✓
 - C) Joining Okazaki fragments
 - D) Adding nucleotides
-

2. DNA gyrase in bacteria is classified as:

- A) Ligase
 - B) Helicase
 - C) Topoisomerase ✓
 - D) Primase
-

3. The main role of topoisomerase is to:

- A) Break RNA
- B) Cut and reseal DNA strands ✓
- C) Translate proteins
- D) Synthesize primers

4. Excessive twisting of DNA during replication is prevented by:

- A) Ligase
 - B) Topoisomerase ✓
 - C) RNA polymerase
 - D) Ribosome
-

5. The semiconservative model of DNA replication was proposed by:

- A) Watson and Crick
 - B) Meselson and Stahl ✓
 - C) Darwin
 - D) Mendel
-

6. In semiconservative replication, each daughter DNA contains:

- A) Two old strands
 - B) Two new strands
 - C) One old and one new strand ✓
 - D) Random fragments
-

7. The conservative model of DNA replication suggests:

- A) DNA is destroyed
 - B) Parental DNA remains intact ✓
 - C) DNA is partially degraded
 - D) RNA replaces DNA
-

8. The dispersive model proposes that DNA replication produces:

- A) Completely new DNA
 - B) Mixed old and new segments ✓
 - C) Only RNA molecules
 - D) Protein-DNA hybrids
-

9. Meselson and Stahl used which isotope in their experiment?

- A) Oxygen-18
 - B) Nitrogen-15 ✓
 - C) Carbon-14
 - D) Hydrogen-3
-

10. The enzyme responsible for extending telomeres is:

- A) DNA polymerase
 - B) Telomerase ✓
 - C) Ligase
 - D) Helicase
-

11. Telomeres are located at:

- A) Middle of chromosome
 - B) End of chromosomes ✓
 - C) Centromere
 - D) RNA strands
-

12. Telomerase contains which component for template activity?

- A) DNA template
 - B) RNA template ✓
 - C) Protein only
 - D) Lipid layer
-

13. The primary function of telomeres is to:

- A) Encode proteins
 - B) Protect chromosome ends ✓
 - C) Form ribosomes
 - D) Unwind DNA
-

14. Telomerase activity is highest in:

- A) Somatic cells
 - B) Germ cells and stem cells ✓
 - C) Mature RBCs
 - D) Bacteria only
-

15. Central dogma describes flow of genetic information as:

- A) Protein → DNA → RNA
 - B) DNA → RNA → Protein ✓
 - C) RNA → DNA → lipid
 - D) DNA → lipid → protein
-

16. The process of DNA making RNA is called:

- A) Translation
- B) Transcription ✓

- C) Replication
 - D) Mutation
-

17. The process of RNA making protein is called:

- A) Transcription
 - B) Translation ✓
 - C) Replication
 - D) Repair
-

18. Reverse transcription is carried out by:

- A) DNA polymerase
 - B) Reverse transcriptase ✓
 - C) Ligase
 - D) Helicase
-

19. Reverse transcription occurs in:

- A) Bacteria
 - B) Retroviruses ✓
 - C) Plants only
 - D) Ribosomes
-

20. Topoisomerase acts on DNA by affecting:

- A) Amino acids
 - B) Supercoiling ✓
 - C) Ribosomes
 - D) Lipids
-

21. DNA replication fork is formed by:

- A) Ligase
 - B) Helicase ✓
 - C) Ribosome
 - D) tRNA
-

22. DNA polymerase synthesizes DNA in the direction:

- A) 3' → 5'
- B) 5' → 3' ✓
- C) Both directions
- D) Random

23. RNA primer is synthesized by:

- A) DNA polymerase
 - B) Primase ✓
 - C) Ligase
 - D) Helicase
-

24. Okazaki fragments are found on:

- A) Leading strand
 - B) Lagging strand ✓
 - C) RNA strand
 - D) Telomeres
-

25. DNA ligase is responsible for:

- A) DNA unwinding
 - B) Joining DNA fragments ✓
 - C) RNA synthesis
 - D) Protein synthesis
-

26. The central dogma was proposed by:

- A) Darwin
 - B) Francis Crick ✓
 - C) Mendel
 - D) Watson only
-

27. DNA gyrase is especially important in:

- A) Humans
 - B) Bacteria ✓
 - C) Plants
 - D) Viruses
-

28. Telomerase prevents loss of DNA during:

- A) Translation
 - B) Replication ✓
 - C) Transcription
 - D) Mutation
-

29. The replication fork moves due to:

- A) Ribosomes
 - B) Helicase activity ✓
 - C) tRNA
 - D) Ligase
-

30. The DNA replication model supported by evidence is:

- A) Conservative
 - B) Semiconservative ✓
 - C) Dispersive
 - D) Random
-

31. DNA topoisomerase I functions by:

- A) Cutting one DNA strand ✓
 - B) Cutting both strands
 - C) Making RNA
 - D) Adding proteins
-

32. Topoisomerase II functions by:

- A) Cutting one strand
 - B) Cutting both strands ✓
 - C) Making RNA primer
 - D) Joining fragments
-

33. Telomerase is structurally a:

- A) Lipid enzyme
 - B) Ribonucleoprotein ✓
 - C) Carbohydrate
 - D) DNA only
-

34. Telomere shortening is associated with:

- A) Cell division ✓
 - B) Protein synthesis
 - C) RNA formation
 - D) Lipid metabolism
-

35. Central dogma does NOT include:

- A) Replication
- B) Transcription

- C) Translation
 - D) Photosynthesis ✓
-

36. DNA replication starts at:

- A) Telomere
 - B) Origin of replication ✓
 - C) Centromere
 - D) RNA primer
-

37. DNA helicase uses energy from:

- A) ATP ✓
 - B) Glucose
 - C) Oxygen
 - D) RNA
-

38. RNA primer provides a:

- A) 5' phosphate
 - B) Free 3' OH group ✓
 - C) Protein base
 - D) Lipid tail
-

39. DNA replication is described as:

- A) Conservative
 - B) Semi-conservative ✓
 - C) Destructive
 - D) Random
-

40. Topoisomerase reduces stress in DNA by:

- A) Adding nucleotides
 - B) Cutting and rejoining DNA ✓
 - C) Making RNA
 - D) Breaking proteins
-

41. The enzyme that removes RNA primer is:

- A) DNA polymerase I ✓
- B) Helicase
- C) Ligase
- D) Primase

42. Telomerase is most active in:

- A) Aging cells
 - B) Cancer cells ✓
 - C) RBCs
 - D) Muscle fibers
-

43. DNA replication ensures:

- A) Genetic continuity ✓
 - B) Protein formation
 - C) RNA destruction
 - D) Lipid synthesis
-

44. Central dogma can be summarized as:

- A) DNA → Protein → RNA
 - B) DNA → RNA → Protein ✓
 - C) RNA → DNA → lipid
 - D) Protein → DNA → RNA
-

45. Topoisomerase prevents:

- A) DNA replication
 - B) DNA tangling ✓
 - C) RNA formation
 - D) Protein synthesis
-

46. Telomerase extends DNA using:

- A) Protein template
 - B) RNA template ✓
 - C) Lipid template
 - D) Sugar template
-

47. Okazaki fragments are joined by:

- A) Ligase ✓
 - B) Helicase
 - C) Primase
 - D) RNA polymerase
-

48. DNA polymerase works on:

- A) Leading strand only
 - B) Both strands ✓
 - C) RNA only
 - D) Protein only
-

49. DNA gyrase is essential for:

- A) Eukaryotic transcription
 - B) Prokaryotic replication ✓
 - C) Translation
 - D) Protein folding
-

50. The central dogma explains:

- A) Energy flow in cells
 - B) Genetic information flow ✓
 - C) Protein folding
 - D) Cell division
-

1. The process of RNA synthesis from a DNA template is called:

- A) Replication
 - B) Transcription ✓
 - C) Translation
 - D) Mutation
-

2. The enzyme responsible for RNA synthesis is:

- A) DNA polymerase
 - B) RNA polymerase ✓
 - C) Ligase
 - D) Helicase
-

3. During transcription, RNA is synthesized in the direction:

- A) $3' \rightarrow 5'$
 - B) $5' \rightarrow 3'$ ✓
 - C) Both directions
 - D) Random
-

4. The strand of DNA used as template during transcription is called:

- A) Coding strand
- B) Template strand ✓

- C) Lagging strand
 - D) Leading strand
-

5. The coding strand of DNA is also called:

- A) Antisense strand
 - B) Non-template strand ✓
 - C) RNA strand
 - D) Primer strand
-

6. RNA polymerase binds to a region called:

- A) Terminator
 - B) Promoter ✓
 - C) Enhancer only
 - D) Telomere
-

7. The promoter region signals the start of:

- A) Translation
 - B) Transcription ✓
 - C) Replication
 - D) Mutation
-

8. The termination of transcription occurs at:

- A) Promoter
 - B) Terminator sequence ✓
 - C) Start codon
 - D) Telomere
-

9. RNA polymerase does NOT require:

- A) DNA template
 - B) RNA primer ✓
 - C) Nucleotides
 - D) Promoter
-

10. The RNA produced during transcription is:

- A) tRNA
- B) mRNA ✓
- C) rRNA only
- D) DNA

11. The genetic code is based on sequences of:

- A) Amino acids
 - B) Codons ✓
 - C) Lipids
 - D) Enzymes
-

12. A codon consists of how many nucleotides?

- A) 1
 - B) 2
 - C) 3 ✓
 - D) 4
-

13. The start codon in most organisms is:

- A) UAA
 - B) AUG ✓
 - C) UGA
 - D) UAG
-

14. Stop codons include:

- A) AUG only
 - B) UAA, UAG, UGA ✓
 - C) AAA only
 - D) GGG only
-

15. The genetic code is degenerate because:

- A) One codon codes many amino acids
 - B) Multiple codons code same amino acid ✓
 - C) It has no stop codons
 - D) It is random
-

16. The genetic code is universal means:

- A) Different in every organism
 - B) Same in most organisms ✓
 - C) Only in humans
 - D) Only in bacteria
-

17. Transcription occurs in eukaryotes in the:

- A) Cytoplasm
 - B) Nucleus ✓
 - C) Ribosome
 - D) Cell membrane
-

18. In prokaryotes, transcription occurs in the:

- A) Nucleus
 - B) Cytoplasm ✓
 - C) Mitochondria only
 - D) Ribosome
-

19. RNA polymerase catalyzes formation of:

- A) Peptide bonds
 - B) Phosphodiester bonds ✓
 - C) Hydrogen bonds
 - D) Glycosidic bonds
-

20. The first step of transcription is:

- A) Elongation
 - B) Initiation ✓
 - C) Termination
 - D) Translation
-

21. The stage where RNA chain is lengthened is:

- A) Initiation
 - B) Elongation ✓
 - C) Termination
 - D) Replication
-

22. The final step of transcription is:

- A) Initiation
 - B) Elongation
 - C) Termination ✓
 - D) Mutation
-

23. RNA polymerase moves along DNA in direction:

- A) 5' → 3' template
- B) 3' → 5' template ✓

- C) Both directions
 - D) Random
-

24. The DNA strand not used as template is:

- A) Template strand
 - B) Coding strand ✓
 - C) Lagging strand
 - D) RNA strand
-

25. The mRNA produced carries information for:

- A) Lipid synthesis
 - B) Protein synthesis ✓
 - C) DNA replication
 - D) Cell wall formation
-

26. RNA synthesis uses which type of nucleotides?

- A) dNTPs
 - B) rNTPs ✓
 - C) Amino acids
 - D) Fatty acids
-

27. RNA polymerase does NOT require:

- A) DNA template
 - B) RNA primer ✓
 - C) NTPs
 - D) Promoter
-

28. The promoter region in bacteria is often called:

- A) TATA box
 - B) -10 and -35 regions ✓
 - C) Telomere
 - D) Codon region
-

29. The TATA box is important for:

- A) Replication
- B) Transcription initiation ✓
- C) Translation
- D) Mutation repair

30. RNA transcript is complementary to:

- A) Coding strand
 - B) Template strand ✓
 - C) Protein
 - D) Ribosome
-

31. The enzyme that unwinds DNA during transcription is:

- A) RNA polymerase ✓
 - B) Ligase
 - C) Helicase only
 - D) Primase
-

32. RNA polymerase does NOT proofread like DNA polymerase:

- A) True ✓
 - B) False
 - C) Only in bacteria
 - D) Only in viruses
-

33. Transcription produces:

- A) DNA
 - B) RNA ✓
 - C) Protein
 - D) Lipid
-

34. The direction of RNA synthesis is determined by:

- A) Ribosome
 - B) DNA template ✓
 - C) tRNA
 - D) ATP
-

35. The genetic code is read during:

- A) Transcription
 - B) Translation ✓
 - C) Replication
 - D) Mutation
-

36. RNA polymerase binds to DNA at:

- A) Enhancer
 - B) Promoter ✓
 - C) Codon
 - D) Stop site
-

37. The enzyme that separates DNA strands during transcription is:

- A) RNA polymerase ✓
 - B) Ligase
 - C) Topoisomerase
 - D) Primase
-

38. The RNA produced is initially called:

- A) Pre-mRNA ✓
 - B) DNA
 - C) Protein
 - D) tDNA
-

39. The genetic code is read in:

- A) Single bases
 - B) Triplets ✓
 - C) Double bases
 - D) Random units
-

40. Each codon specifies:

- A) One amino acid ✓
 - B) Many proteins
 - C) One gene
 - D) One chromosome
-

41. RNA polymerase requires which ion for activity?

- A) Ca^{2+}
 - B) Mg^{2+} ✓
 - C) Na^{+}
 - D) Cl^{-}
-

42. Transcription stops when RNA polymerase reaches:

- A) Promoter
- B) Terminator ✓

- C) Codon
 - D) Ribosome
-

43. The strand that determines RNA sequence is:

- A) Coding strand
 - B) Template strand ✓
 - C) Lagging strand
 - D) Protein strand
-

44. RNA synthesis is a part of:

- A) Central dogma ✓
 - B) Photosynthesis
 - C) Respiration
 - D) Digestion
-

45. RNA polymerase II synthesizes:

- A) tRNA
 - B) mRNA ✓
 - C) rRNA only
 - D) DNA
-

46. rRNA is mainly involved in:

- A) Protein synthesis ✓
 - B) DNA replication
 - C) Lipid formation
 - D) Mutation
-

47. tRNA function is to:

- A) Carry amino acids ✓
 - B) Make DNA
 - C) Break proteins
 - D) Store genes
-

48. Genetic code ensures correct:

- A) DNA replication
- B) Amino acid sequence ✓
- C) Lipid formation
- D) Cell division

49. Transcription is regulated by:

- A) Ribosomes
 - B) Transcription factors ✓
 - C) tRNA
 - D) Lysosomes
-

50. RNA polymerase synthesizes RNA without needing:

- A) DNA template
 - B) Primer ✓
 - C) NTPs
 - D) Enzyme
-

CONCEPTUAL Q/A LEC 1 - 104

1. What is DNA replication?

DNA replication is the biological process in which DNA makes an exact copy of itself during cell division.

It ensures genetic continuity.

It occurs in the S phase of the cell cycle.

2. Why is DNA replication called semiconservative?

Because each new DNA molecule contains one parental strand and one newly synthesized strand.

This was proven by Meselson and Stahl.

It ensures accuracy of genetic transfer.

3. What is the function of DNA helicase?

It unwinds the DNA double helix.

It breaks hydrogen bonds between strands.

It forms the replication fork.

4. What is a replication fork?

It is the Y-shaped region where DNA unwinds.

Replication machinery assembles here.

It moves along the DNA during replication.

5. What is the role of DNA polymerase?

It synthesizes new DNA strands.

It adds nucleotides in 5'→3' direction.

It also proofreads DNA.

6. Why is RNA primer required in DNA replication?

DNA polymerase cannot start synthesis alone.

RNA primer provides free 3'-OH group.

It is synthesized by primase.

7. What is primase?

Primase is an enzyme that synthesizes RNA primer.

It initiates DNA replication.

It works at the replication fork.

8. What are Okazaki fragments?

They are short DNA segments on lagging strand.

They are synthesized discontinuously.

They are later joined by ligase.

9. What is the function of DNA ligase?

It joins Okazaki fragments.

It forms phosphodiester bonds.

It seals gaps in DNA strands.

10. Why is lagging strand discontinuous?

Because DNA polymerase works only 5'→3'.

Lagging strand runs opposite direction.

So it is made in fragments.

11. What is leading strand?

It is synthesized continuously during replication.

It moves toward replication fork.

It requires only one primer.

12. What is DNA topoisomerase?

It relieves supercoiling in DNA.
It cuts and rejoins DNA strands.
It prevents DNA tangling.

13. What is DNA gyrase?

It is a bacterial topoisomerase.
It introduces negative supercoils.
It helps in replication.

14. Why is topoisomerase important?

It prevents DNA overwinding.
It reduces torsional stress.
It allows smooth replication.

15. What is RNA structure?

RNA is usually single-stranded.
It contains ribose sugar and uracil.
It plays role in protein synthesis.

16. How is RNA different from DNA?

RNA has ribose and uracil.
DNA has deoxyribose and thymine.
RNA is usually single-stranded.

17. What is transcription?

It is synthesis of RNA from DNA.
It is the first step of gene expression.
It is catalyzed by RNA polymerase.

18. What is RNA polymerase?

It is enzyme that synthesizes RNA.
It binds to promoter region.
It does not need a primer.

19. What is promoter region?

It is DNA sequence where transcription starts.

RNA polymerase binds here.
It controls gene expression.

20. What is terminator sequence?

It signals end of transcription.
RNA polymerase stops here.
RNA transcript is released.

21. What is coding strand?

It is DNA strand not used as template.
It has same sequence as RNA (except T instead of U).
It is also called sense strand.

22. What is template strand?

It is DNA strand used for RNA synthesis.
RNA is complementary to it.
It is antisense strand.

23. What is genetic code?

It is set of rules for protein synthesis.
It uses codons of 3 bases.
It is universal and degenerate.

24. What is a codon?

A codon is three nucleotide sequence.
It codes for amino acid.
It is read during translation.

25. What is start codon?

It signals start of protein synthesis.
Usually AUG.
It codes for methionine.

26. What are stop codons?

They signal termination of translation.
UAA, UAG, UGA.
They do not code amino acids.

27. What is degeneracy of genetic code?

Multiple codons can code same amino acid.
It reduces mutation effects.
It increases genetic stability.

28. What is universality of genetic code?

Same codons code same amino acids in most organisms.
It shows evolutionary relationship.
Rare exceptions exist in mitochondria.

29. What is central dogma?

Flow of genetic information in cells.
DNA → RNA → Protein.
Explains gene expression.

30. What is transcription in central dogma?

Process of making RNA from DNA.
Occurs in nucleus (eukaryotes).
Uses RNA polymerase.

31. What is translation?

Process of protein synthesis from mRNA.
Occurs at ribosomes.
Uses tRNA and rRNA.

32. What is reverse transcription?

RNA makes DNA.
Occurs in retroviruses.
Uses reverse transcriptase.

33. What is telomere?

It is repetitive DNA at chromosome ends.
Protects chromosomes from damage.
Shortens during replication.

34. What is telomerase?

It is enzyme that extends telomeres.
Uses RNA template.
Active in stem and cancer cells.

35. Why do telomeres shorten?

Because lagging strand cannot fully replicate ends.
RNA primers are removed.
This leads to gradual shortening.

36. What is replication model?

It explains how DNA is copied.
Includes semiconservative, conservative, dispersive models.
Only semiconservative is correct.

37. What is conservative model?

Parental DNA remains intact.
New DNA is completely new.
Not experimentally supported.

38. What is dispersive model?

Old and new DNA are mixed in fragments.
Each strand is patchwork.
Not accepted today.

39. Who proved DNA replication model?

Meselson and Stahl.
Used nitrogen isotope experiment.
Confirmed semiconservative replication.

40. What is role of ATP in helicase?

Provides energy for unwinding DNA.
Helps break hydrogen bonds.
Essential for replication fork.

41. What is single-strand binding protein?

It stabilizes separated DNA strands.

Prevents re-annealing.
Keeps DNA open for replication.

42. What is phosphodiester bond?

Bond between nucleotides in DNA/RNA.
Formed by polymerases.
Creates backbone of nucleic acids.

43. What is direction of DNA synthesis?

Always 5' to 3'.
Template read 3' to 5'.
Universal rule of polymerases.

44. Why is DNA replication accurate?

Due to proofreading by DNA polymerase.
Corrects errors during synthesis.
Maintains genetic stability.

45. What is mutation?

Change in DNA sequence.
May occur during replication.
Can be harmful or beneficial.

46. What is origin of replication?

Site where DNA replication begins.
Proteins assemble here.
Multiple in eukaryotes.

47. What is bidirectional replication?

Replication occurs in two directions.
Two replication forks form.
Speeds up DNA copying.

48. What is lagging strand problem?

DNA polymerase works only 5' → 3'.
Opposite strand needs fragments.
Solved by Okazaki fragments.

49. What is RNA primer removal enzyme?

DNA polymerase I.
Replaces RNA with DNA.
Works in prokaryotes.

50. What is significance of DNA replication?

Ensures genetic inheritance.
Enables cell division.
Maintains species continuity.

51. What is the main product of transcription?

Transcription produces RNA from a DNA template.
The main product is mRNA.
It carries genetic information for protein synthesis.

52. Why is RNA synthesis called transcription?

Because genetic information is copied from DNA to RNA.
No change in information occurs, only rewriting.
It is the first step of gene expression.

53. What is the role of RNA polymerase in transcription?

It catalyzes RNA synthesis.
It binds to promoter region.
It forms phosphodiester bonds between nucleotides.

54. Why does RNA polymerase not need a primer?

It can start RNA synthesis de novo.
It directly adds first nucleotide.
Unlike DNA polymerase, no 3'-OH primer is required.

55. What is the function of promoter in transcription?

It signals the start of transcription.
It provides binding site for RNA polymerase.
It regulates gene expression.

56. What is the role of terminator sequence?

It signals end of transcription.
RNA polymerase detaches from DNA.
RNA transcript is released.

57. What is template strand in transcription?

It is DNA strand used for RNA synthesis.
RNA is complementary to it.
Also called antisense strand.

58. What is coding strand?

It is DNA strand not used as template.
Its sequence matches RNA (except T replaced by U).
Also called sense strand.

59. What is mRNA function?

It carries genetic code from DNA to ribosome.
It serves as template for protein synthesis.
It determines amino acid sequence.

60. What is tRNA function?

It carries amino acids to ribosome.
It matches codons with anticodons.
It acts as adapter molecule.

61. What is rRNA function?

It forms ribosome structure.
It catalyzes peptide bond formation.
It is essential for translation.

62. What is transcription unit?

Segment of DNA transcribed into RNA.
Includes promoter, coding region, terminator.
Defines gene expression unit.

63. What is RNA polymerase II responsible for?

It synthesizes mRNA in eukaryotes.

It is essential for protein coding genes.
It works in nucleus.

64. What is RNA polymerase I responsible for?

It synthesizes rRNA.
It functions in nucleolus.
Important for ribosome formation.

65. What is RNA polymerase III responsible for?

It synthesizes tRNA and small RNAs.
It supports protein synthesis indirectly.
Works in nucleus.

66. What is initiation in transcription?

RNA polymerase binds to promoter.
DNA strands begin to unwind.
First RNA nucleotides are added.

67. What is elongation in transcription?

RNA strand grows longer.
RNA polymerase moves along DNA.
Nucleotides are added continuously.

68. What is termination in transcription?

RNA synthesis stops.
RNA polymerase detaches.
RNA transcript is released.

69. What is transcription bubble?

Region where DNA strands are separated.
Site of active RNA synthesis.
Moves along DNA during transcription.

70. What is genetic code?

Set of rules that converts nucleotide sequence into amino acids.
Read in triplets (codons).
Universal and specific.

71. Why is genetic code triplet?

Because 3 bases encode one amino acid.
Provides enough combinations for 20 amino acids.
Ensures coding efficiency.

72. What is a codon?

A sequence of three nucleotides in mRNA.
Codes for specific amino acid or stop signal.
Read during translation.

73. What is anticodon?

Three-base sequence in tRNA.
Complementary to mRNA codon.
Ensures correct amino acid placement.

74. What is start codon function?

Initiates translation process.
Codes for methionine (AUG).
Sets reading frame.

75. What is stop codon function?

Signals termination of protein synthesis.
Does not code for amino acid.
Releases polypeptide chain.

76. What are stop codons?

UAA, UAG, UGA.
They end translation process.
Recognized by release factors.

77. Why is genetic code degenerate?

Multiple codons can code same amino acid.
Reduces harmful effects of mutations.
Provides genetic stability.

78. What is universality of genetic code?

Same codons code same amino acids in most organisms.
Indicates common ancestry.
Few exceptions exist in mitochondria.

79. What is wobble hypothesis?

Flexible base pairing at third codon position.
Explains degeneracy of genetic code.
Reduces number of tRNAs needed.

80. What is transcription factor?

Protein that regulates transcription.
Helps RNA polymerase bind DNA.
Controls gene expression.

81. What is gene expression?

Process of converting gene information into functional product.
Includes transcription and translation.
Controls cellular function.

82. Why is DNA not directly used for protein synthesis?

Because DNA is too stable and protected in nucleus.
RNA acts as messenger.
Ensures controlled expression.

83. What is primary transcript?

Initial RNA produced from DNA.
Contains introns and exons.
Requires processing in eukaryotes.

84. What is RNA processing?

Modification of pre-mRNA.
Includes splicing, capping, poly-A tail addition.
Occurs in nucleus.

85. What is splicing?

Removal of introns from pre-mRNA.

Joining of exons.
Produces mature mRNA.

86. What is 5' capping?

Addition of modified guanine at 5' end.
Protects mRNA from degradation.
Helps ribosome binding.

87. What is poly-A tail?

Chain of adenine nucleotides at 3' end.
Increases mRNA stability.
Helps in export from nucleus.

88. What is intron?

Non-coding region in gene.
Removed during RNA processing.
Does not code for protein.

89. What is exon?

Coding region of gene.
Expressed in final mRNA.
Translated into protein.

90. What is transcription regulation?

Control of gene expression at RNA level.
Involves repressors and activators.
Ensures cell-specific protein production.

91. What is operon system?

Group of genes regulated together in bacteria.
Controlled by single promoter.
Example: lac operon.

92. What is lac operon?

Gene system for lactose metabolism.
Induced when lactose is present.
Regulates enzyme production.

93. Why is transcription important?

It transfers genetic information from DNA.
It initiates protein synthesis.
It controls cell activity.

94. What is role of magnesium in transcription?

Acts as cofactor for RNA polymerase.
Stabilizes enzyme structure.
Essential for catalysis.

95. What is antisense RNA?

RNA complementary to mRNA.
Blocks gene expression.
Used in regulation.

96. What is ribozyme?

RNA molecule with enzymatic activity.
Can catalyze reactions.
Important in RNA world hypothesis.

97. What is RNA world hypothesis?

Theory that RNA was first genetic material.
RNA acted as both enzyme and genetic code.
Preceded DNA evolution.

98. What is difference between transcription and replication?

Transcription produces RNA from DNA.
Replication produces DNA from DNA.
Transcription is selective, replication is complete.

99. Why is transcription selective?

Only specific genes are transcribed.
Controlled by regulatory proteins.
Allows cell specialization.

100. What is significance of transcription?

It is first step of gene expression.
Produces RNA for protein synthesis.
Controls biological functions.

1. What are biomolecules?

Biomolecules are molecules found in living organisms.
They include carbohydrates, proteins, lipids, and nucleic acids.
They are essential for life processes.

2. Why are biomolecules important?

They build cell structure and perform functions.
They store and transfer energy.
They regulate biological activities.

3. What are macromolecules?

Large complex biomolecules.
Formed by polymerization of smaller units.
Examples include proteins and nucleic acids.

4. What are monomers?

Basic building blocks of polymers.
They join to form macromolecules.
Example: glucose, amino acids.

5. What are carbohydrates?

Organic compounds made of carbon, hydrogen, oxygen.
They are primary energy sources.
General formula is $(CH_2O)_n$.

6. What is the main function of carbohydrates?

Provide energy to cells.
Store energy in plants and animals.
Form structural components.

7. What are monosaccharides?

Simple sugars.

Single sugar units.

Example: glucose, fructose.

8. What is glucose?

A monosaccharide.

Primary energy source for cells.

Used in respiration.

9. What are disaccharides?

Formed by two monosaccharides.

Joined by glycosidic bond.

Example: sucrose, lactose.

10. What are polysaccharides?

Long chains of monosaccharides.

Used for storage or structure.

Example: starch, cellulose.

11. What is starch?

Storage carbohydrate in plants.

Made of glucose units.

Digested to release energy.

12. What is glycogen?

Storage carbohydrate in animals.

Stored in liver and muscles.

Highly branched structure.

13. What is cellulose?

Structural carbohydrate in plants.

Forms plant cell walls.

Indigestible for humans.

14. What is chitin?

Structural polysaccharide.

Found in exoskeleton of insects.
Also in fungal cell walls.

15. What is glycosidic bond?

Bond between two sugar molecules.
Forms carbohydrates.
Created by dehydration reaction.

16. Why are carbohydrates called sugars?

Because many are sweet-tasting monosaccharides.
They dissolve easily in water.
Provide quick energy.

17. What is dehydration synthesis?

Process of joining monomers by removing water.
Forms glycosidic bonds.
Important in polymer formation.

18. What is hydrolysis?

Breakdown of polymers using water.
Opposite of dehydration synthesis.
Used in digestion.

19. Why is glucose important in respiration?

It is primary fuel for ATP production.
Broken down in glycolysis.
Provides energy for cells.

20. What is classification of carbohydrates based on size?

Monosaccharides, disaccharides, polysaccharides.
Based on number of sugar units.
Determines function.

21. What is ribose?

A five-carbon sugar.
Found in RNA.
Important in nucleotides.

22. What is deoxyribose?

Sugar in DNA.

Lacks one oxygen atom compared to ribose.

Provides DNA stability.

23. Why are carbohydrates hydrophilic?

They contain many hydroxyl groups.

They form hydrogen bonds with water.

They dissolve easily.

24. What is energy storage role of carbohydrates?

They store glucose in polymers.

Starch in plants and glycogen in animals.

Used when energy is needed.

25. What is structural role of carbohydrates?

They provide rigidity to cells.

Cellulose in plants.

Chitin in fungi and insects.

26. What is oligosaccharide?

Short chain of monosaccharides.

Between mono and polysaccharides.

Found on cell surfaces.

27. What are glycoproteins?

Proteins with carbohydrate groups.

Involved in cell recognition.

Important in immune response.

28. What are glycolipids?

Lipids with carbohydrate chains.

Found in cell membranes.

Help in cell signaling.

29. Why is cellulose strong?

Due to beta-1,4 glycosidic bonds.
Forms long straight chains.
Creates strong fibers.

30. Why humans cannot digest cellulose?

We lack enzyme cellulase.
Beta bonds cannot be broken.
Acts as dietary fiber.

31. What is starch digestion?

Broken down by amylase enzyme.
Produces glucose.
Provides energy.

32. What is reducing sugar?

Sugar that can donate electrons.
Has free aldehyde or ketone group.
Example: glucose.

33. What is non-reducing sugar?

Cannot reduce other molecules.
No free aldehyde group.
Example: sucrose.

34. Why is sucrose non-reducing?

Because both anomeric carbons are bonded.
No free reactive group.
Prevents oxidation reaction.

35. What is carbohydrate metabolism?

Breakdown and synthesis of sugars.
Provides energy and storage.
Controlled by enzymes.

36. What is ATP relationship with carbohydrates?

Carbohydrates produce ATP.

ATP stores energy.
Central energy currency.

37. What is glycogenesis?

Formation of glycogen from glucose.
Occurs in liver and muscles.
Energy storage process.

38. What is glycogenolysis?

Breakdown of glycogen.
Releases glucose.
Used during energy demand.

39. What is photosynthesis role in carbohydrates?

Produces glucose in plants.
Uses sunlight energy.
Base of food chain.

40. What is cellulose microfibril?

Bundles of cellulose chains.
Provide structural strength.
Found in plant cell walls.

41. Why are carbohydrates polar molecules?

Due to hydroxyl groups.
They form hydrogen bonds.
Increase solubility.

42. What is carbon skeleton in carbohydrates?

Chain of carbon atoms.
Forms backbone of sugars.
Determines structure.

43. What is triose sugar?

Sugar with 3 carbon atoms.
Example: glyceraldehyde.
Intermediate in metabolism.

44. What is pentose sugar?

Sugar with 5 carbons.

Example: ribose and deoxyribose.

Important in nucleic acids.

45. What is hexose sugar?

Sugar with 6 carbon atoms.

Example: glucose.

Main energy source.

46. What is isomerism in carbohydrates?

Same formula, different structure.

Affects function.

Example: glucose and fructose.

47. What is alpha glucose?

Form of glucose with OH group down.

Used in starch.

Easily digested.

48. What is beta glucose?

Form of glucose with OH group up.

Forms cellulose.

Structural role.

49. Why is glucose highly important?

It is universal energy source.

Used in respiration.

Found in all organisms.

50. What is carbohydrate classification basis?

Based on number of sugar units.

Also based on functional groups.

Determines biological role.

51. Why are lipids considered efficient energy storage molecules compared to carbohydrates?

Because lipids are more reduced molecules.

They release more ATP per gram.

They are stored without water, increasing energy density.

52. Why do lipids not dissolve in water even though they contain oxygen?

Because their structure is dominated by nonpolar hydrocarbon chains.

Oxygen presence is too low to create polarity.

So they remain hydrophobic.

53. What is the biological advantage of lipid insolubility?

It allows long-term energy storage without affecting osmotic balance.

Prevents leakage in cells.

Also enables membrane formation.

54. Why are fats better storage molecules than glycogen for long-term energy?

Fats are more compact and store more energy per unit mass.

They do not bind water.

Glycogen is bulky and hydrated.

55. Why is fat storage preferred in animals rather than carbohydrate storage?

Because animals need lightweight energy reserves.

Fat provides more energy per gram.

It avoids excessive body weight from water-bound glycogen.

56. What makes triglycerides chemically stable molecules?

They contain ester bonds with low reactivity.

Hydrophobic chains reduce interaction with water.

This increases storage stability.

57. Why is cholesterol important despite being a lipid?

Because it regulates membrane fluidity.

It stabilizes cell membranes at different temperatures.

Also acts as precursor for hormones.

58. Why can cholesterol both increase and decrease membrane fluidity?

At high temperature it restricts movement of phospholipids.

At low temperature it prevents tight packing.

Thus it maintains balance.

59. What makes phospholipids ideal for membrane formation?

They are amphipathic molecules.

Have hydrophilic heads and hydrophobic tails.

They naturally form bilayers.

60. Why do phospholipids spontaneously form bilayers in water?

Because hydrophobic effect drives tail avoidance of water.

Heads interact with water.

This self-assembly minimizes energy.

61. What is the biological significance of lipid bilayer fluidity?

It allows movement of proteins and lipids.

Supports membrane transport.

Enables cell signaling.

62. Why are membranes not rigid structures?

Because lipid molecules are not fixed.

They move laterally within bilayer.

This provides flexibility.

63. What role do unsaturated fatty acids play in membrane fluidity?

They introduce kinks in hydrocarbon chains.

Prevent tight packing.

Increase membrane fluidity.

64. Why are saturated fats solid at room temperature?

Because they pack tightly in straight chains.

Strong van der Waals forces stabilize them.

This increases melting point.

65. Why are unsaturated fats usually liquid?

Because double bonds create bends.

Weakens molecular packing.
Reduces melting point.

66. What makes lipids suitable for insulation in organisms?

They conduct heat poorly.
They are stored under skin.
They reduce heat loss.

67. Why are lipids used in hormone synthesis?

Because they are structurally flexible.
Can be modified into signaling molecules.
Steroid hormones are lipid-derived.

68. What is the advantage of steroid hormones being lipid-based?

They can easily pass through membranes.
No need for transport proteins.
They act inside cells.

69. Why do lipid-soluble substances pass easily through membranes?

Because membrane core is hydrophobic.
Lipids dissolve in lipid bilayer.
No energy required.

70. Why are fatty acids important in metabolism?

They are broken into acetyl-CoA.
Used in Krebs cycle.
Produce large ATP amounts.

71. Why is beta-oxidation important?

It breaks fatty acids into energy units.
Produces acetyl-CoA.
Occurs in mitochondria.

72. Why do lipids store more energy than carbohydrates chemically?

They contain more hydrogen atoms.
More oxidation occurs during breakdown.
Thus more ATP is produced.

73. Why are fats hydrophobic but not completely non-reactive?

They still contain ester bonds.
But hydrocarbon chains are nonpolar.
So reactivity is limited.

74. Why are waxes highly protective biologically?

They are extremely hydrophobic.
Prevent water loss.
Used in plant cuticles and animal coverings.

75. Why do plants use wax coatings on leaves?

To reduce water loss by evaporation.
To protect from pathogens.
To maintain internal water balance.

76. Why are lipids important in cell signaling?

Some lipids act as second messengers.
They regulate cellular responses.
Example: phosphoinositides.

77. Why is lipid metabolism tightly regulated?

Because excess lipids can cause diseases.
Balance is needed for energy homeostasis.
Controlled by hormones.

78. Why is adipose tissue important?

It stores energy as fat.
Provides insulation and cushioning.
Acts as endocrine organ.

79. Why is fat digestion slower than carbohydrate digestion?

Because lipids are hydrophobic.
Require emulsification by bile salts.
Enzymatic breakdown is slower.

80. Why is bile important in lipid digestion?

It emulsifies fats into smaller droplets.
Increases surface area for enzymes.
Does not chemically digest fats.

81. Why are lipases required for fat digestion?

Because fats cannot be directly absorbed.
Lipases break triglycerides into fatty acids.
They enable absorption.

82. Why do lipids form droplets in water?

Because they are hydrophobic.
They aggregate to minimize water contact.
This reduces energy.

83. Why are lipids not considered polymers like proteins?

They are not made of repeating monomers.
They are assembled from different components.
So they are not true polymers.

84. Why are lipids called amphipathic in membranes?

Because they have both polar and nonpolar regions.
Heads interact with water.
Tails avoid water.

85. Why is membrane asymmetry important?

Different lipids are present in each layer.
This supports signaling and transport.
It maintains cell function.

86. Why do cells regulate cholesterol levels?

Too much increases rigidity.
Too little destabilizes membrane.
Balance is essential.

87. Why are lipid rafts important?

They organize membrane proteins.

Help in signaling pathways.
Provide functional microdomains.

88. Why do lipids not form linear structures like carbohydrates?

Because they are non-polymeric and branched.
They lack repeating monomers.
They are structurally diverse.

89. Why are triglycerides hydrophobic overall?

Because ester groups are masked by fatty acid chains.
Nonpolar region dominates.
Water interaction is minimal.

90. Why do animals store fat instead of starch?

Fat provides more energy density.
It is lighter for movement.
Starch is mainly plant storage.

91. Why is lipid oxidation important in fasting?

Because glucose reserves are limited.
Body switches to fat breakdown.
Provides sustained energy.

92. Why are phospholipids critical for life?

They form the basis of cell membranes.
Create selective barriers.
Enable compartmentalization.

93. Why is compartmentalization important in cells?

It separates biochemical reactions.
Increases efficiency.
Prevents interference between pathways.

94. Why do lipids contribute to buoyancy in aquatic organisms?

Because they are less dense than water.
Reduce overall body density.
Help floating.

95. Why are lipid membranes selectively permeable?

Hydrophobic core restricts polar molecules.
Only small nonpolar molecules pass easily.
Transport proteins regulate others.

96. Why is lipid oxidation more energy-rich than carbohydrate oxidation?

Because lipids are more reduced.
They release more electrons.
Generate more ATP.

97. Why do phospholipids never dissolve completely in water?

Because only heads interact with water.
Tails remain hydrophobic.
This creates bilayer formation.

98. Why is lipid storage preferred in seeds?

Because it provides concentrated energy.
Supports germination.
Occupies less space.

99. Why are lipids important in brain function?

They form myelin sheath.
Improve nerve impulse conduction.
Provide structural support.

100. Why is lipid metabolism linked to hormones?

Because hormones regulate fat breakdown and synthesis.
Insulin and glucagon control balance.
Maintains energy homeostasis.

101. Why are proteins considered the most functionally diverse biomolecules?

Because they can form enzymes, structural components, and signaling molecules.
Their function depends on 3D structure.
Sequence variation creates huge diversity.

102. Why does protein function depend on its shape?

Because shape determines binding ability.
Active sites fit specific substrates.
Even small changes can disrupt function.

103. What happens if a protein loses its shape?

It becomes non-functional.
This process is called denaturation.
Biological activity is lost.

104. Why is denaturation often irreversible?

Because peptide bonds remain but folding interactions break.
Refolding is difficult under normal conditions.
Structure collapses permanently.

105. Why are enzymes highly specific?

Because active sites have precise shapes.
Only specific substrates fit.
This follows lock-and-key or induced fit model.

106. What is the significance of the induced fit model?

It explains enzyme flexibility.
Active site changes shape upon binding.
Improves catalytic efficiency.

107. Why do enzymes lower activation energy?

They stabilize transition state.
Reduce energy barrier for reactions.
Speed up biochemical processes.

108. Why are enzymes not consumed in reactions?

Because they act as catalysts.
They are regenerated after reaction.
Only facilitate reaction speed.

109. Why is temperature important for enzyme activity?

It affects molecular motion.

High temperature increases collisions but may denature enzymes.
Low temperature slows reaction.

110. Why does extreme temperature reduce enzyme activity?

Because protein structure is disrupted.
Active site changes shape.
Substrate binding is reduced.

111. Why is pH important for enzyme function?

It affects ionization of amino acids.
Changes active site structure.
Each enzyme has optimum pH.

112. Why are enzymes called biological catalysts?

Because they speed up reactions in living systems.
They are protein-based.
They are highly efficient.

113. Why are proteins polymers?

Because they are made of amino acid monomers.
Joined by peptide bonds.
Form long chains.

114. What is a peptide bond?

A bond between amino acids.
Formed by dehydration reaction.
Creates protein backbone.

115. Why is protein folding important?

Because function depends on 3D structure.
Correct folding creates active sites.
Misfolding causes disease.

116. What are the levels of protein structure?

Primary, secondary, tertiary, quaternary.
Each level adds complexity.
Determines function.

117. What is primary structure of protein?

Sequence of amino acids.
Determined by DNA.
Forms protein backbone.

118. What is secondary structure?

Local folding patterns like alpha helices and beta sheets.
Stabilized by hydrogen bonds.
Gives structural shape.

119. What is tertiary structure?

3D folding of entire polypeptide.
Stabilized by multiple interactions.
Creates functional protein.

120. What is quaternary structure?

Combination of multiple polypeptide chains.
Forms functional protein complexes.
Example: hemoglobin.

121. Why is hemoglobin considered a quaternary protein?

Because it has four subunits.
Each binds oxygen.
Works cooperatively.

122. Why is oxygen binding in hemoglobin cooperative?

Binding of one oxygen increases affinity for others.
Changes protein shape.
Improves oxygen transport.

123. Why are amino acids called amphoteric?

They can act as acid and base.
Contain amino and carboxyl groups.
Maintain pH balance.

124. What determines amino acid properties?

Side chain (R group).

It can be polar, nonpolar, charged.

Affects protein behavior.

125. Why are some amino acids hydrophobic?

Because they have nonpolar side chains.

Avoid water interaction.

Found in protein core.

126. Why are some amino acids hydrophilic?

Because they have polar or charged groups.

Interact with water.

Found on protein surface.

127. Why is enzyme specificity important in metabolism?

Prevents unwanted reactions.

Ensures controlled pathways.

Maintains efficiency.

128. Why do enzymes require cofactors?

Some reactions need additional molecules.

Cofactors assist enzyme function.

Improve catalytic activity.

129. What are coenzymes?

Organic cofactors.

Often derived from vitamins.

Assist enzyme reactions.

130. Why are vitamins important for enzymes?

They act as coenzymes.

Support metabolic reactions.

Deficiency affects enzymes.

131. Why is protein synthesis essential for life?

Proteins perform almost all cell functions.

Include enzymes and structural proteins.
Without them, life cannot exist.

132. Why do mutations affect proteins?

Because DNA sequence determines amino acids.
Changes alter protein structure.
May affect function.

133. What is enzyme-substrate complex?

Temporary structure formed during reaction.
Substrate binds active site.
Reaction occurs here.

134. Why is active site specific?

Because amino acid arrangement is precise.
Creates binding pocket.
Ensures selectivity.

135. Why are proteins sensitive to environmental changes?

Because weak bonds maintain structure.
Heat or pH disrupts interactions.
Leads to denaturation.

136. Why is collagen strong?

Because of triple helix structure.
Rich in glycine and proline.
Provides tensile strength.

137. Why is keratin important?

It provides structural support.
Found in hair and nails.
Resistant to damage.

138. Why is enzyme inhibition important?

Regulates metabolic pathways.
Controls reaction rates.
Prevents overproduction.

139. What is competitive inhibition?

Inhibitor competes with substrate.
Binds active site.
Reduces enzyme activity.

140. What is non-competitive inhibition?

Inhibitor binds elsewhere.
Changes enzyme shape.
Reduces activity.

141. Why do enzymes have optimal temperature?

Because structure is stable at certain conditions.
Too high causes denaturation.
Too low reduces activity.

142. Why is protein diversity so high?

Because 20 amino acids can combine in many ways.
Sequence variations are vast.
Leads to functional diversity.

143. Why is protein synthesis tightly regulated?

To prevent waste of energy.
Ensures correct cell function.
Maintains balance.

144. Why is insulin a protein hormone?

Because peptide structure allows signaling.
It regulates glucose metabolism.
Acts on target cells.

145. Why are antibodies proteins?

Because they need specificity.
Protein structure allows binding diversity.
Important in immune defense.

146. Why is enzyme activity graph bell-shaped for pH?

Because extreme pH disrupts structure.
Only optimal pH maintains shape.
Activity decreases on both sides.

147. Why are enzymes reusable?

Because they are not consumed.
Remain unchanged after reaction.
Catalyze multiple cycles.

148. Why is ATP linked to proteins?

Because many protein functions require energy.
ATP powers conformational changes.
Drives active processes.

149. Why is protein misfolding dangerous?

It leads to loss of function.
Can form toxic aggregates.
Associated with diseases.

150. Why are proteins central to cell function?

Because they perform structural, enzymatic, and regulatory roles.
They control almost all biological processes.
Life depends on them.

SUMMARIZED NOTES LEC 1 - 104

Cell biomolecules are the fundamental chemical substances that build and control living organisms. They include carbohydrates, lipids, proteins, and nucleic acids, all of which are essential for structure, energy storage, and biological regulation. Biomolecules exist as monomers (small units like glucose and amino acids) and macromolecules (large polymers like proteins and DNA), where monomers are linked through chemical bonds such as glycosidic bonds in carbohydrates and peptide bonds in proteins. Their structure directly determines their function, meaning even small changes in molecular arrangement can significantly affect biological activity.

Carbohydrates are primarily energy-providing molecules composed of carbon, hydrogen, and oxygen, usually in a 1:2:1 ratio. They exist as monosaccharides like glucose, disaccharides like sucrose, and polysaccharides like starch, glycogen, cellulose, and chitin. They serve roles in quick energy supply, energy storage, and

structural support in plants and animals. Lipids are hydrophobic molecules that are highly efficient energy storage compounds, providing more energy per gram than carbohydrates. They include triglycerides, phospholipids, and steroids. Lipids form cell membranes, act as hormones, provide insulation, and serve in long-term energy storage. Their hydrophobic nature allows them to form stable structures like membranes without dissolving in water.

Proteins are the most diverse biomolecules, made of amino acids linked by peptide bonds. Their function depends on their structure, which is organized into primary, secondary, tertiary, and quaternary levels. Proteins act as enzymes, hormones, structural materials, and transport molecules. Enzymes are biological catalysts that speed up reactions by lowering activation energy and work through specific active sites. Their activity is influenced by temperature, pH, and cofactors, and they can be inhibited through competitive or non-competitive mechanisms.

Nucleic acids include DNA and RNA, which store and transmit genetic information. DNA is double-stranded and contains deoxyribose sugar, while RNA is usually single-stranded and contains ribose. DNA stores genetic instructions, while RNA carries and expresses them through transcription and translation. The central dogma explains the flow of genetic information as DNA → RNA → Protein, which is essential for gene expression. DNA replication follows a semiconservative model where each new DNA molecule contains one old and one new strand, and involves enzymes like helicase, topoisomerase, primase, DNA polymerase, and ligase.

Chromosomes are highly organized structures of DNA and proteins (histones) found in the nucleus. DNA is packed into chromatin, which condenses into chromosomes during cell division. Telomeres are protective ends of chromosomes that shorten during replication, while telomerase helps extend them in specific cells. Overall, biomolecules work together in a highly coordinated system that maintains life, supports metabolism, ensures genetic continuity, and regulates all cellular processes.

DNA, RNA, replication, and gene expression together explain how genetic information is stored, copied, and expressed in living organisms. DNA is the permanent genetic material that stores instructions in the form of nucleotide sequences, while RNA acts as a temporary messenger that carries and expresses this information. DNA replication ensures genetic continuity before cell division by producing two identical DNA molecules through a semiconservative mechanism, where each daughter DNA contains one parental and one newly synthesized strand. This process depends on enzymes such as helicase (unwinding DNA), topoisomerase (removing supercoiling), primase (forming RNA primers), DNA polymerase (adding nucleotides), and ligase (joining fragments).

The central dogma explains the directional flow of genetic information as DNA → RNA → Protein. This means genetic information is first transcribed into RNA and then translated into proteins, which perform most cellular functions. Transcription is the process in which RNA polymerase synthesizes RNA from a DNA template by binding to promoter regions, elongating the RNA strand in the 5'→3' direction, and stopping at terminator sequences. The resulting RNA, mainly mRNA, carries coded information in the form of codons, where each codon consists of three nucleotides that specify one amino acid.

Translation occurs on ribosomes, where mRNA is decoded into a polypeptide chain with the help of tRNA and rRNA. tRNA brings specific amino acids according to anticodon–codon pairing, while rRNA forms the structural and catalytic core of the ribosome. The genetic code is universal, nearly identical in all organisms, and degenerate, meaning multiple codons can code for the same amino acid, which increases genetic stability.

The fluid mosaic model explains the structure of the cell membrane, describing it as a dynamic phospholipid bilayer with embedded proteins, cholesterol, and carbohydrates. Lipids provide fluidity, proteins perform transport and signaling functions, and carbohydrates assist in recognition. The membrane is selectively permeable, allowing controlled movement of substances in and out of the cell, which is essential for homeostasis.

Together, these processes form a continuous biological system: DNA stores information, replication preserves it, transcription converts it into RNA, translation produces proteins, and the fluid mosaic membrane regulates cellular interactions and transport, ensuring proper functioning and survival of the cell.

BIO5105 – MODEL PAPER

SECTION A: MCQs ($14 \times 1 = 14$ Marks)

1. DNA replication is called semiconservative because:

- A) Both strands are newly synthesized
- B) One parental strand is conserved in each daughter DNA ✓
- C) DNA is partially destroyed
- D) RNA replaces one strand

2. Why is DNA helicase essential in replication?

- A) It joins DNA fragments
- B) It stabilizes DNA strands
- C) It separates double helix for access ✓
- D) It synthesizes RNA primer

3. RNA contains uracil instead of thymine because:

- A) It increases DNA stability
- B) It reduces mutation rate
- C) It reflects structural difference in nucleic acid chemistry ✓
- D) It prevents replication

4. DNA polymerase cannot initiate synthesis because:

- A) It lacks energy
 - B) It requires a free 3'-OH group ✓
 - C) It cannot bind DNA
 - D) It works in reverse direction
-

5. RNA polymerase binds specifically to promoter regions because:

- A) They contain stop signals
 - B) They define gene boundaries for transcription initiation ✓
 - C) They stop replication
 - D) They form proteins
-

6. The lagging strand is discontinuous because:

- A) DNA is circular
 - B) Polymerase directionality conflicts with fork movement ✓
 - C) RNA is absent
 - D) Ligase is inactive
-

7. Genetic code is considered degenerate because:

- A) It changes randomly
 - B) Multiple codons encode same amino acid ✓
 - C) It is unstable
 - D) It is non-functional
-

8. Why is the central dogma considered directional?

- A) Because protein controls DNA
 - B) Because information flows DNA → RNA → Protein ✓
 - C) Because RNA controls DNA
 - D) Because proteins store genes
-

9. Why does topoisomerase act ahead of replication fork?

- A) To synthesize RNA
 - B) To prevent supercoiling stress ✓
 - C) To form proteins
 - D) To join fragments
-

10. Telomerase is important because it:

- A) Breaks DNA ends
 - B) Extends chromosome ends to prevent information loss ✓
 - C) Synthesizes proteins
 - D) Stops replication
-

11. Why is transcription considered selective?

- A) Entire genome is always copied
 - B) Only specific genes are expressed ✓
 - C) DNA is fully degraded
 - D) RNA is always active
-

12. Why is phospholipid bilayer stable in water?

- A) Hydrophilic tails repel water
 - B) Hydrophobic effect drives self-assembly ✓
 - C) DNA stabilizes it
 - D) Proteins dissolve it
-

13. RNA primer is required because:

- A) DNA polymerase cannot start synthesis alone ✓
 - B) RNA is genetic material
 - C) It stops replication
 - D) It breaks DNA
-

14. Why is DNA replication highly accurate?

- A) Random enzyme activity
 - B) Proofreading by DNA polymerase ✓
 - C) RNA interference
 - D) Lipid involvement
-

SECTION B: SHORT QUESTIONS (2 × 5 = 10 Marks)

(Conceptual reasoning answers)

Q1. Why is DNA replication considered a highly regulated and error-controlled process?

DNA replication is tightly regulated because genetic information must be preserved accurately for survival.

Enzymes like DNA polymerase not only synthesize DNA but also correct mismatched bases through proofreading.

Multiple enzymes coordinate at the replication fork to prevent errors and structural stress.

This ensures faithful inheritance of genetic information across generations.

Q2. Why does RNA act as an intermediate molecule between DNA and protein?

DNA is stable and protected in the nucleus, so it is not directly used for protein synthesis.

RNA provides a flexible and temporary copy of genetic information.

mRNA carries coded instructions, tRNA interprets codons, and rRNA forms ribosomes.

This separation allows controlled and efficient gene expression.

SECTION C: LONG QUESTIONS (2 × 13 = 26 Marks)

(Deep conceptual + analytical answers)

Q1. Explain how the coordination of enzymes ensures accurate DNA replication at the molecular level.

DNA replication is a highly coordinated enzymatic process ensuring genetic continuity.

Helicase first unwinds the double helix, creating tension ahead of the replication fork.

Topoisomerase relieves this stress by cutting and rejoining DNA strands, preventing tangling.

Primase synthesizes RNA primers because DNA polymerase cannot start synthesis independently.

DNA polymerase extends the new strand in 5'→3' direction while continuously proofreading errors.

On the lagging strand, discontinuous synthesis creates Okazaki fragments which are later joined by ligase.

This enzyme coordination ensures speed, accuracy, and stability of the replication process.

Q2. Explain how the central dogma connects transcription, translation, and genetic code in regulating cellular function.

The central dogma describes the directional flow of genetic information: DNA → RNA → Protein.

During transcription, RNA polymerase converts DNA information into mRNA by reading the template strand.

This mRNA contains codons, which are triplet nucleotide sequences forming the genetic code.

During translation, ribosomes decode codons while tRNA brings specific amino acids via anticodon matching.

Peptide bonds form, producing a polypeptide that folds into a functional protein.

The genetic code is universal and degenerate, ensuring stability and evolutionary conservation.

This integrated system ensures that genetic information is accurately expressed into functional cellular proteins.

VULMS Mastery